

Fundamentals of Solid State Physics

Semiconductors - General

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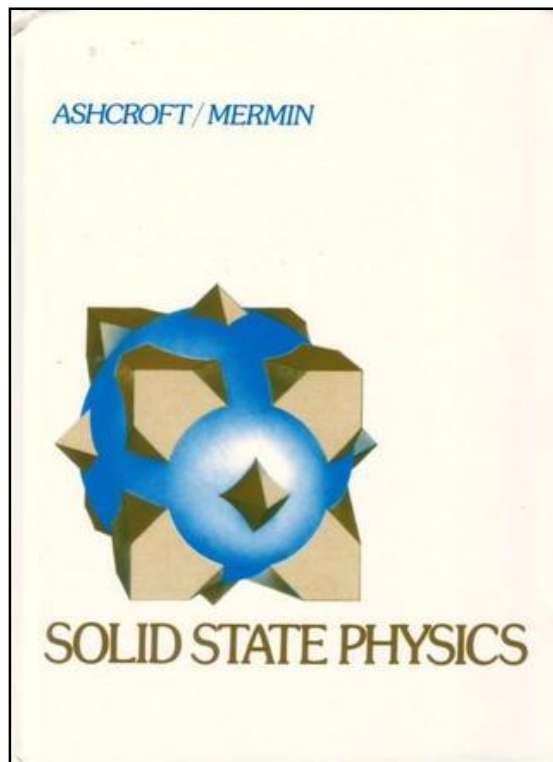


This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Further Reading

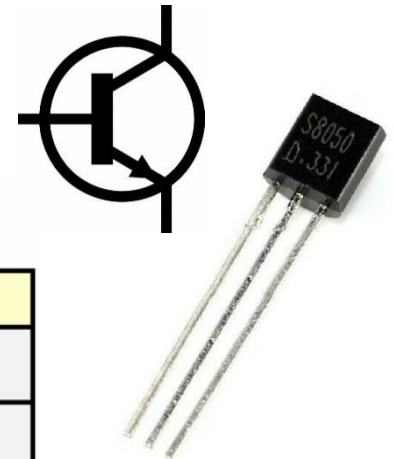
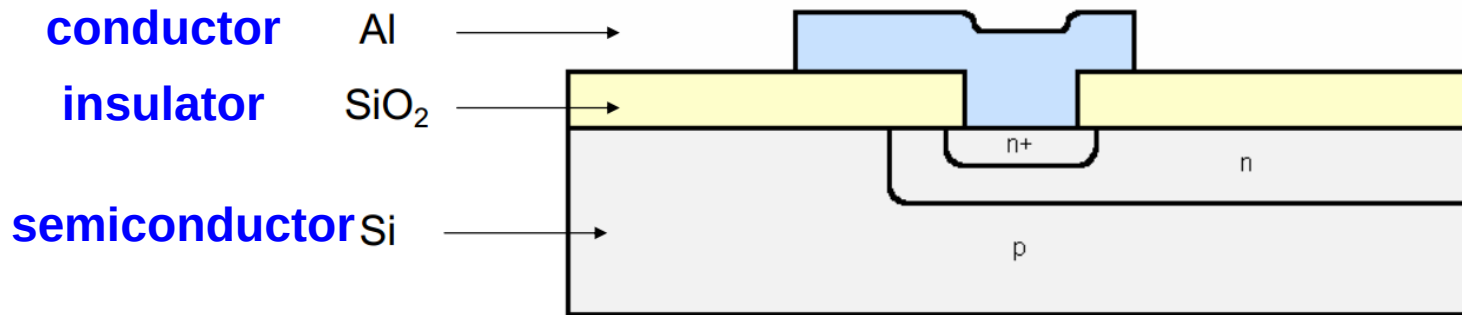
- Ashcroft & Mermin, Chapter 28
- Solid State Electronic Devices by Streetman, Chap.3



Electronic Properties of Materials

CMOS transistor

- Complementary **Metal-Oxide-Semiconductor**



Metal

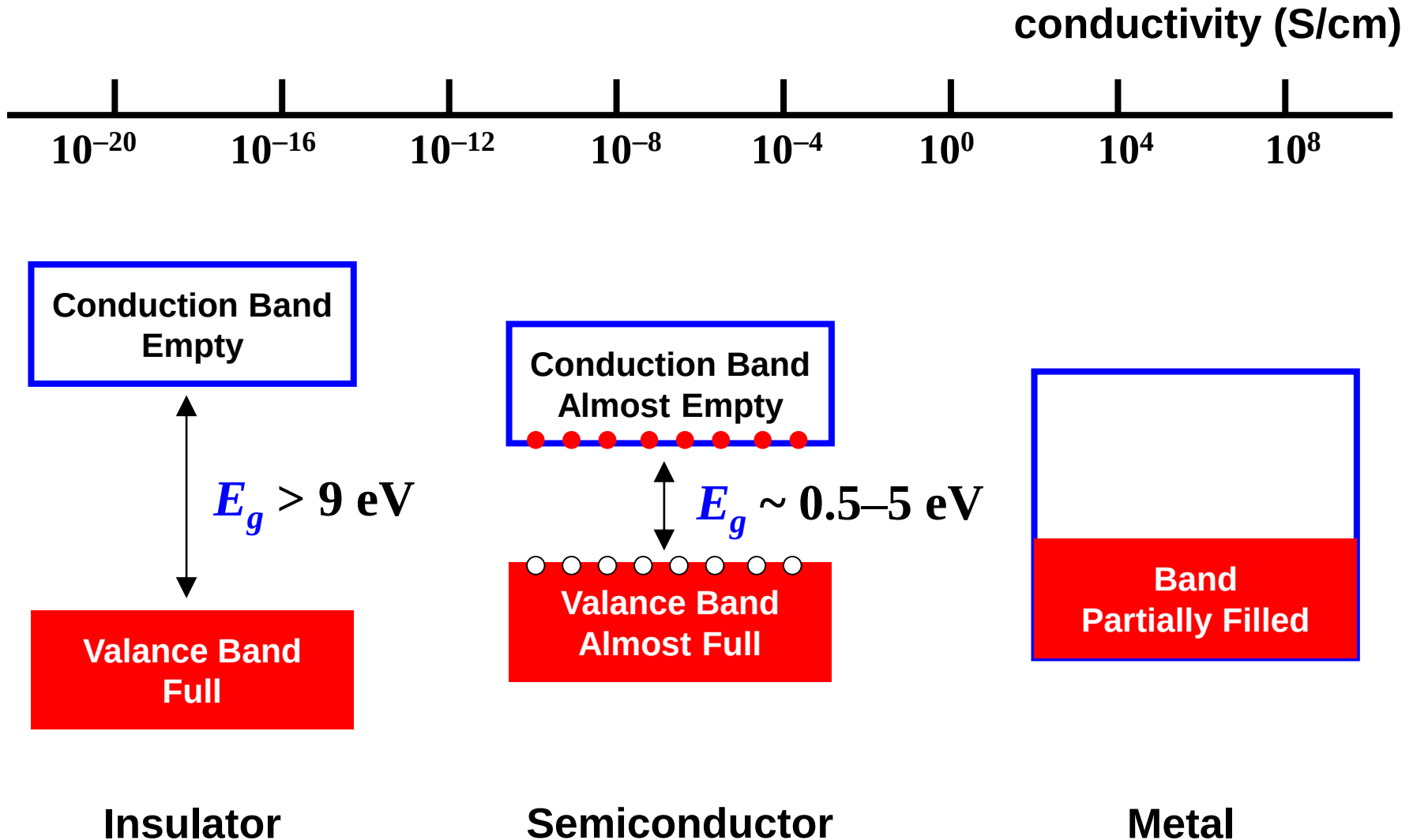


SiO_2



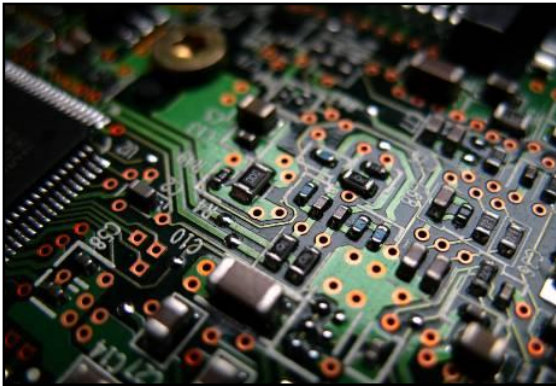
Silicon

Metal, Insulator, Semiconductor



Semiconductors - Applications

semiconductors are the basis of electronics and photonics



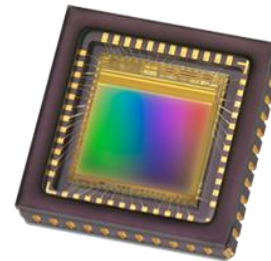
integrated circuits



LEDs



lasers



detectors

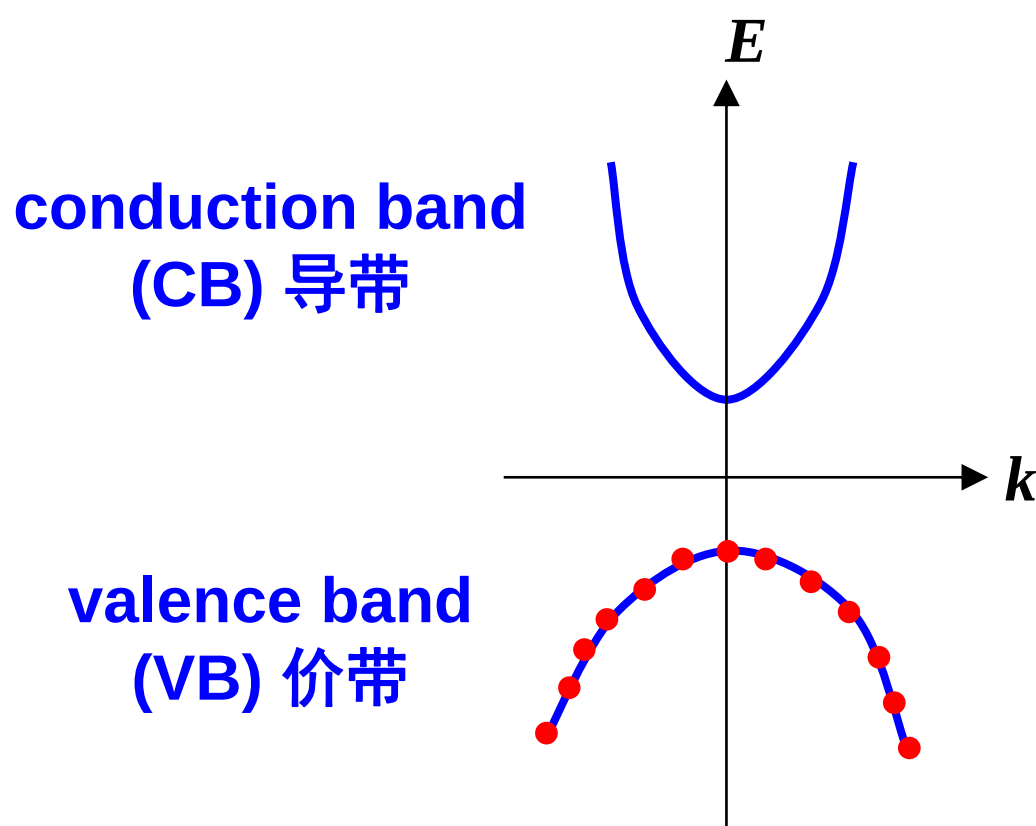


solar cells

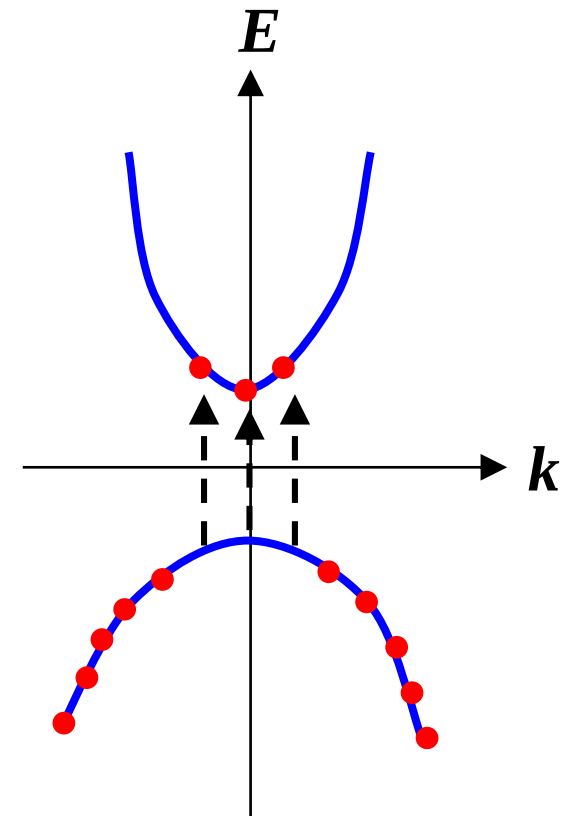
Semiconductors - General Concepts

- Band diagram $E(k)$
- Band gap E_g
- Effective mass m^*
- Holes
- Density of States (DOS) $g(E)$
- Density of Carriers n_c and p_v
 - ▣ Mass Action Law
- Intrinsic and Extrinsic

Semiconductor 半导体



$T = 0 \text{ K}$
CB is empty, VB is full
insulator



$T > 0 \text{ K}$
thermalization 热激发
CB and VB are partly filled
conductor

Band Structure / Diagram 能带图

Free electrons

energy

$$E(k) = \frac{\hbar^2 k^2}{2m_0}$$

velocity

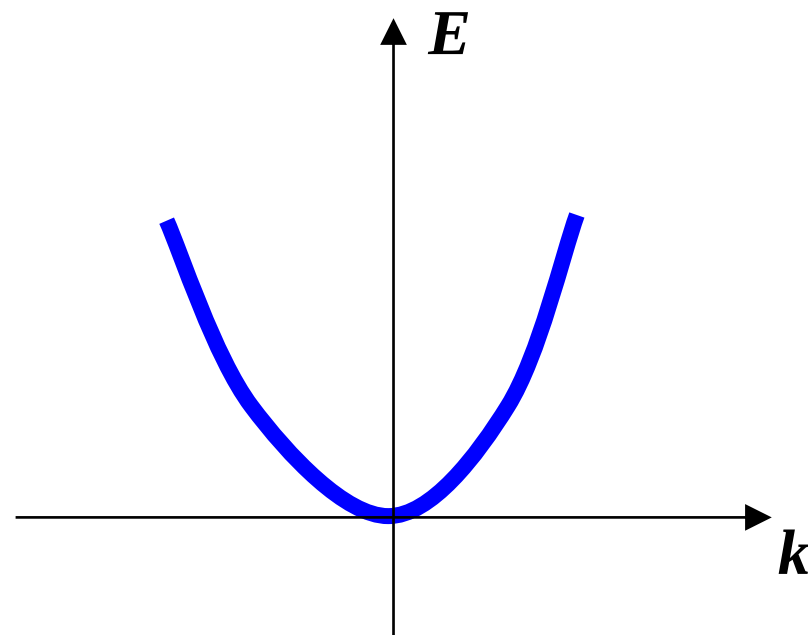
$$v = \frac{\hbar k}{m_0} = \frac{1}{\hbar} \frac{dE(k)}{dk}$$

momentum

$$p = \hbar k$$

electron mass

$$\frac{1}{m_0} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$



E-k diagram
(energy dispersion curve)

Band Structure / Diagram 能带图

energy

$$E(k)$$

band gap

$$E_g$$

crystal momentum
(*not* electron momentum)

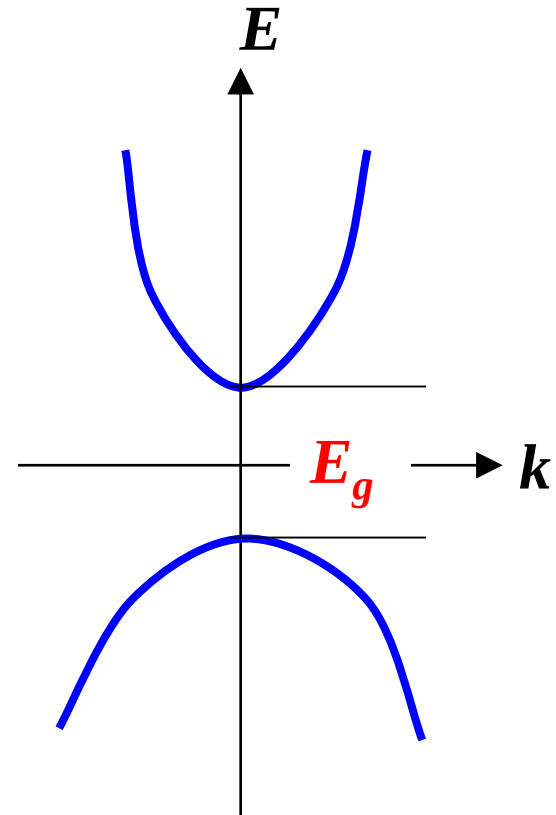
$$\hbar k$$

group velocity

$$v_g = \frac{1}{\hbar} \frac{dE(k)}{dk}$$

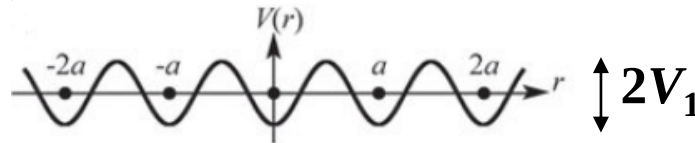
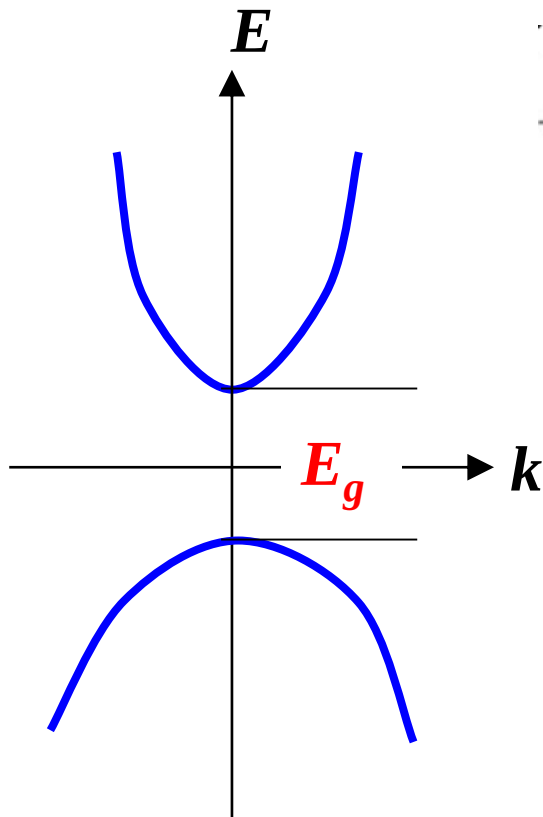
or

$$\mathbf{v} = \frac{1}{\hbar} \nabla_{\mathbf{k}} E(\mathbf{k})$$



E-k diagram
(energy dispersion curve)

Band Gap E_g



$$E_g = 2V_1$$

the nearly free
electron model

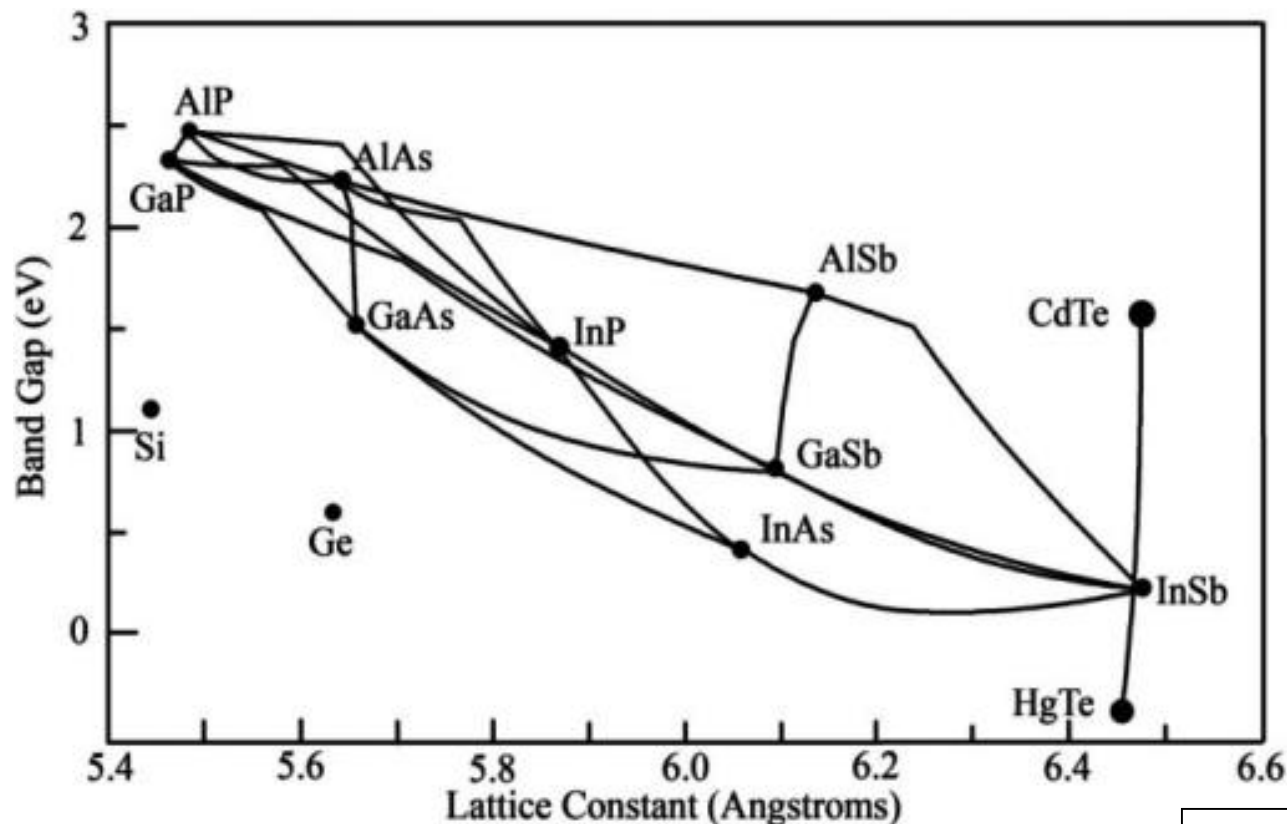
III IV V					2
					He
5 B	6 C	7 N	8 O	9 F	10 Ne
13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn

at $T = 300$ K

	a (Å)	E_g (eV)
C (diamond)	3.57	5.5
Si	5.43	1.1
Ge	5.66	0.66
α -Sn	6.49	0.08

Q: Why?

Band Gap E_g



III IV V					2
					He
5 B	6 C	7 N	8 O	9 F	10 Ne
13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn

Si > Ge

AlAs > GaAs > InAs

GaP > GaAs > GaSb

larger atoms
-> smaller V_1
-> smaller E_g

Band Gap E_g

	E_g (eV)
Si	1.1
AlP	2.5
MgS	4.0
NaCl	8.5

	E_g (eV)
Ge	0.7
GaAs	1.4
ZnSe	3.5
KBr	7.5

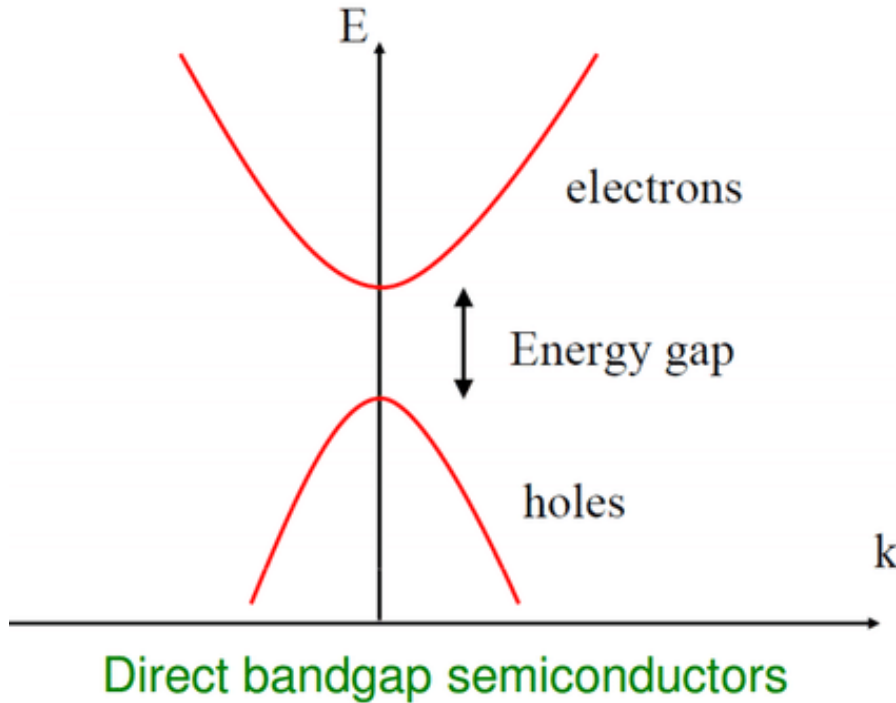
		13 III A 3A	14 IV A 4A	15 V A 5A	16 VI A 6A	17 VII A 7A
		5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 16.000	9 F Fluorine 18.998
Na	Mg	13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453
K	Zn	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904
	Cd	49 In Indium 114.818	50 Sn Tin 118.710	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904

Si < AlP < MgS < NaCl

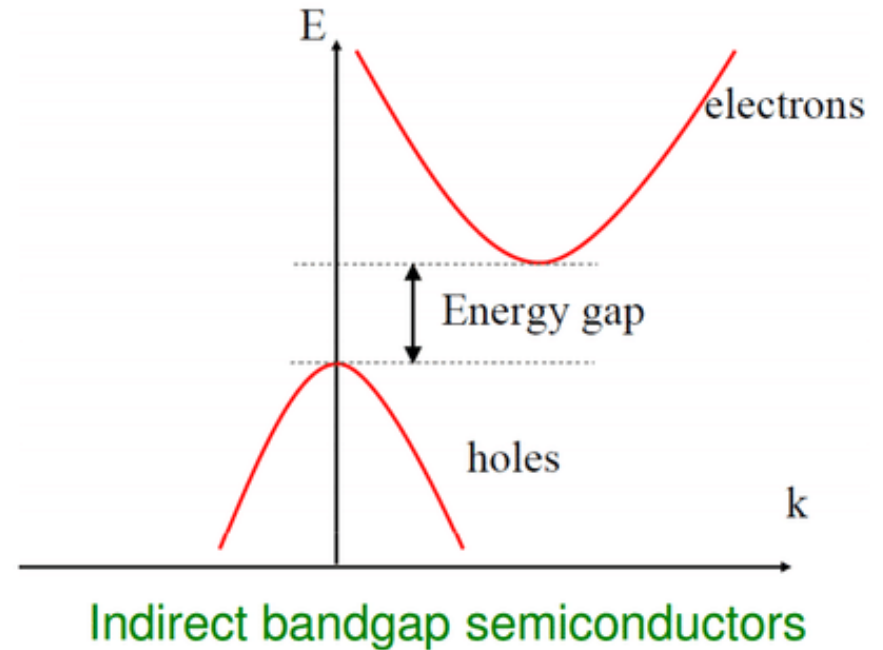
Ge < GaAs < ZnSe < KBr

*more polarity
-> larger V_1
-> larger E_g*

Direct and Indirect Gaps

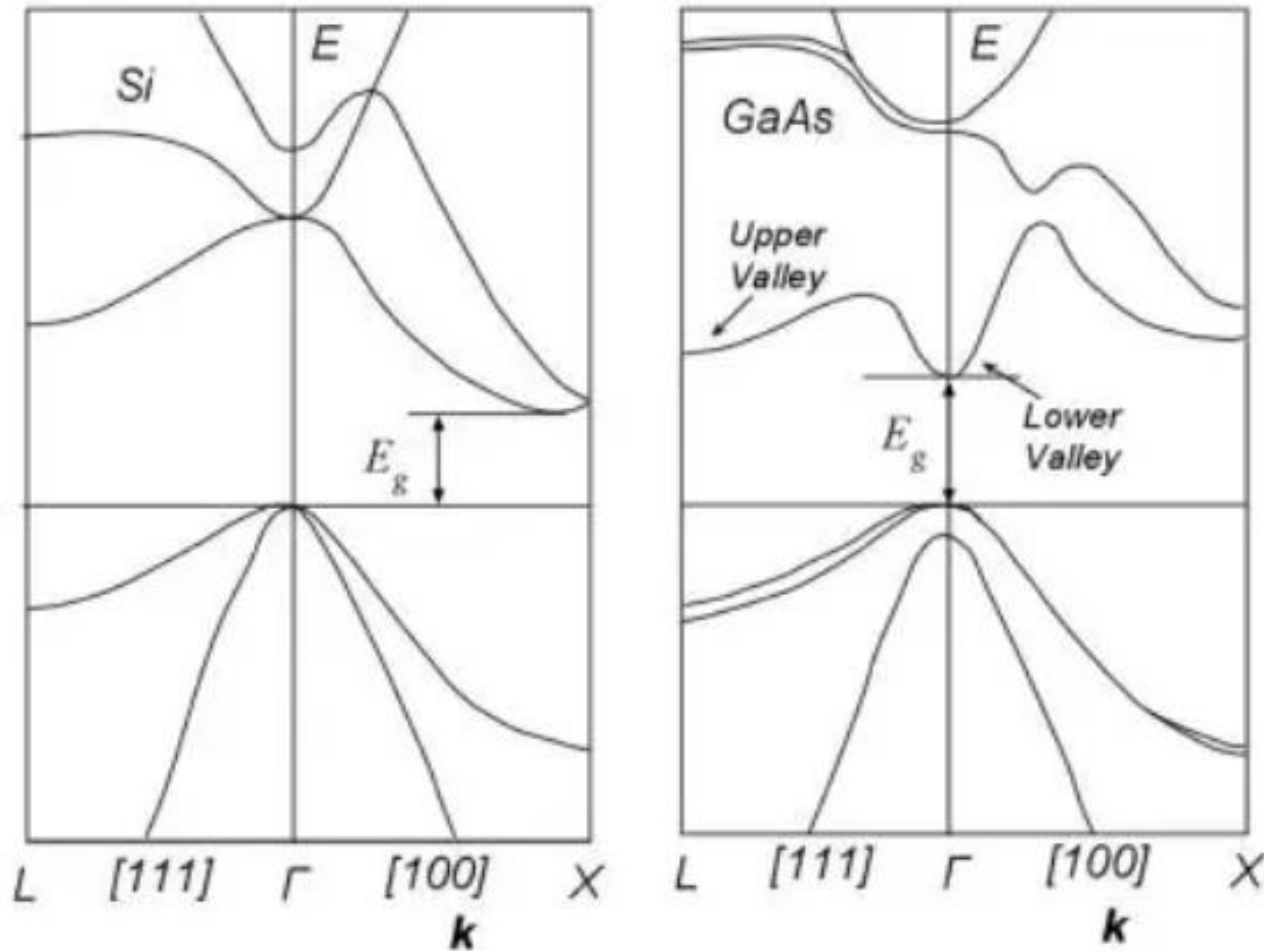


直接帶隙



间接帶隙

Direct and Indirect Gaps



Silicon - indirect

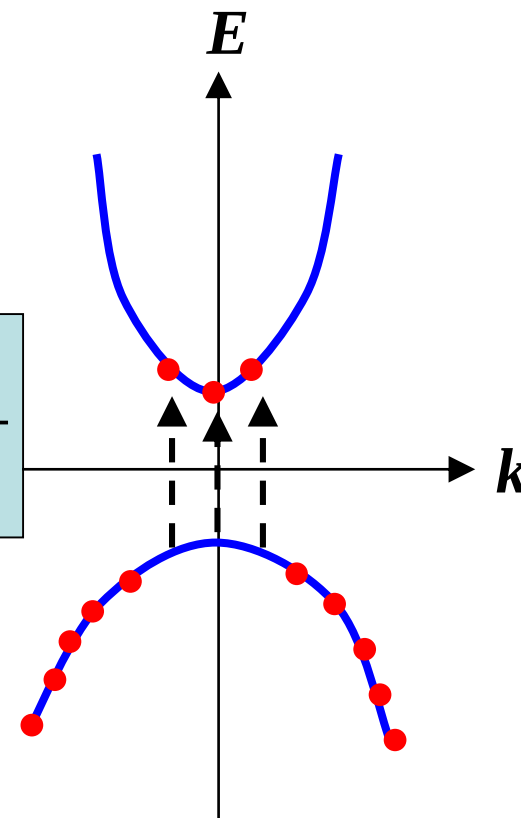
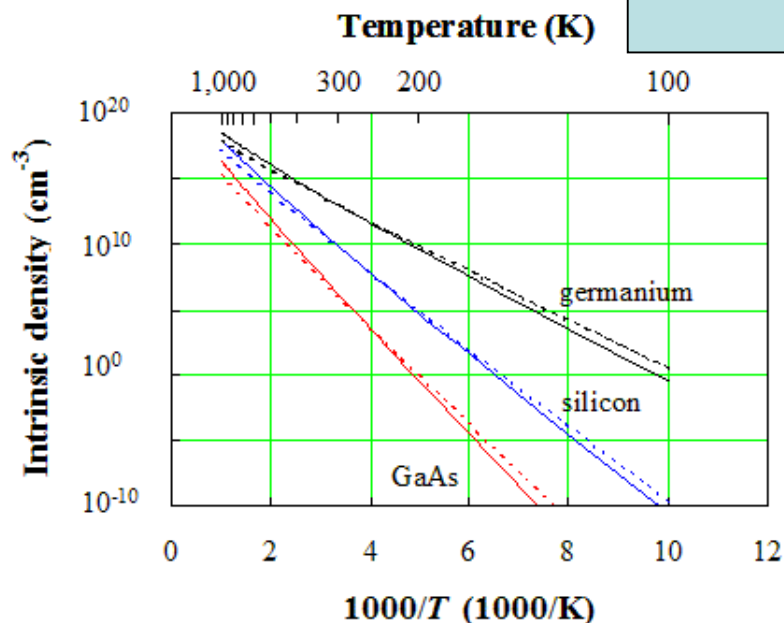
GaAs - direct

Measurement of Band Gaps

temperature dependence
of carrier concentration (or conductivity)

$$\sigma \propto n_i \propto T^{3/2} \cdot e^{-E_g/2k_B T}$$

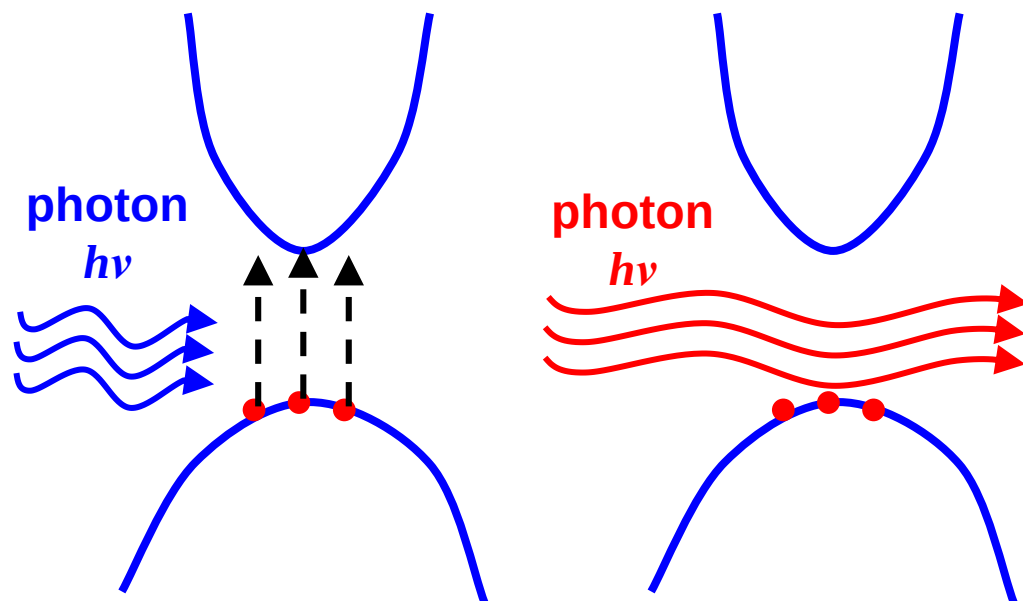
$$\ln \sigma \sim \ln n_i \sim -\frac{E_g}{2k_B T}$$



$T > 0 \text{ K}$
thermalization 热激发
CB and VB are partly filled
conductor

Measurement of Band Gaps

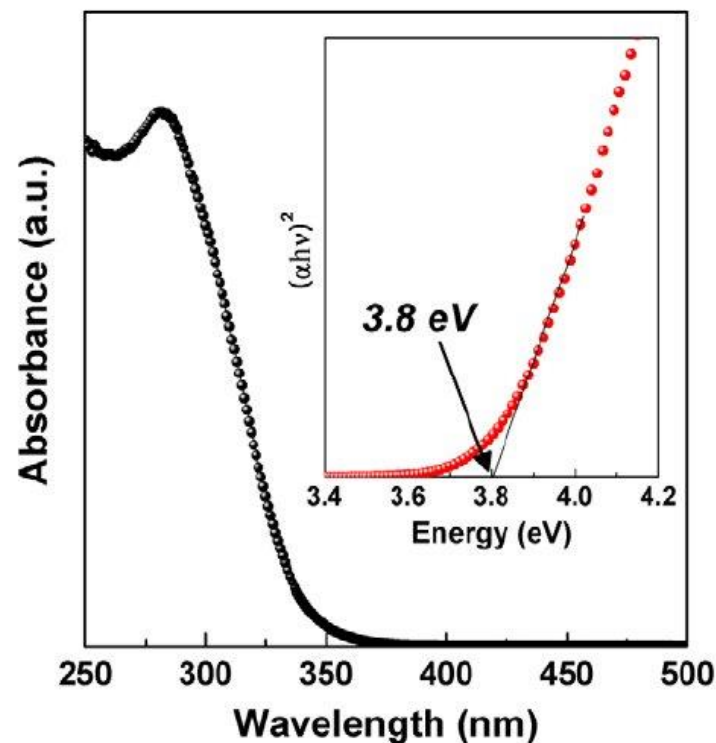
wavelength dependent
optical absorption



$h\nu > E_g$ ---> light absorption

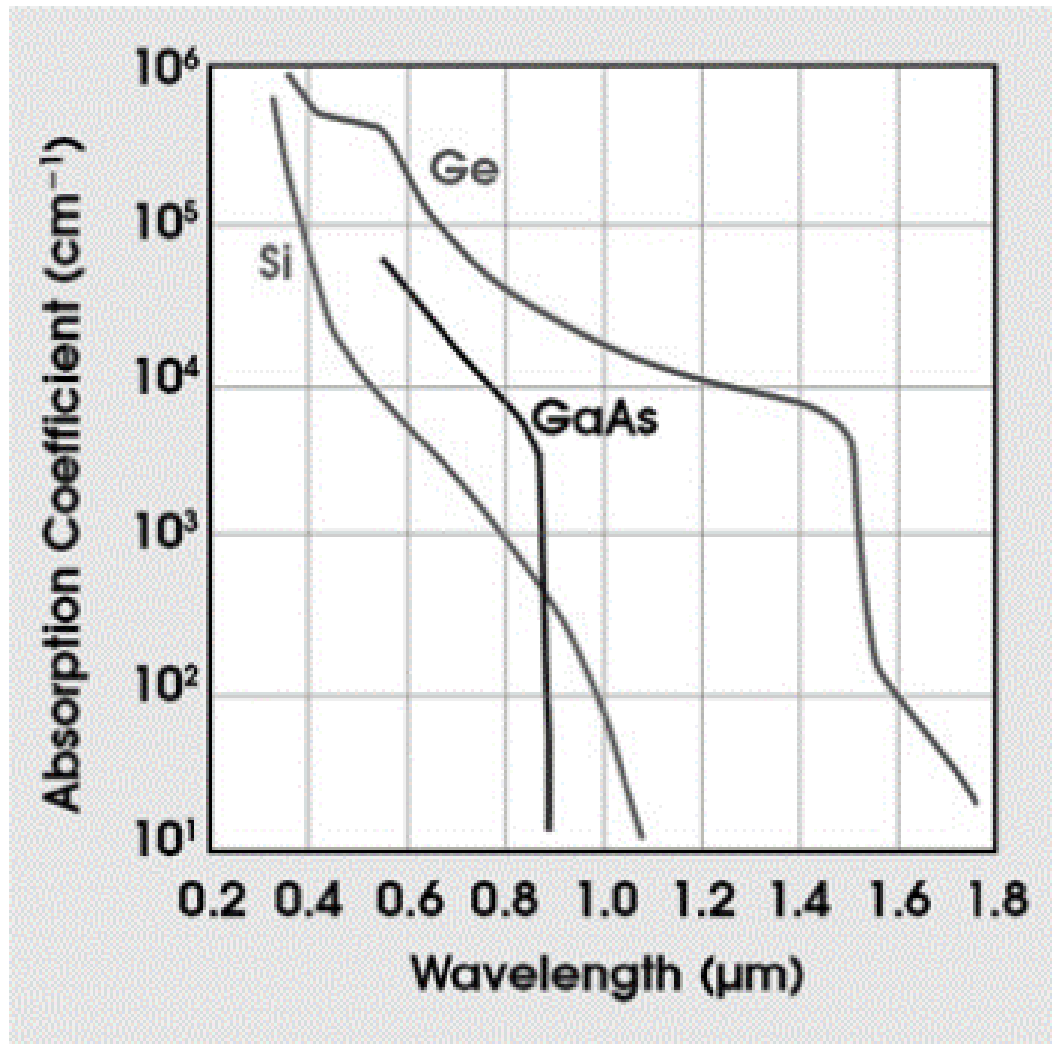
$h\nu < E_g$ ---> light transmission

Example:
 Zn_2SnO_4 nanoparticles



D. Kim, et al, *Nanoscale* 4, 557 (2011)

Measurement of Band Gaps



$$E_g = \frac{hc}{\lambda_g}$$



$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

	E_g (eV)
Si	1.1
Ge	0.66
GaAs	1.43
glass	>3.5

silicon is black

glass is transparent

Effective Mass 有效质量

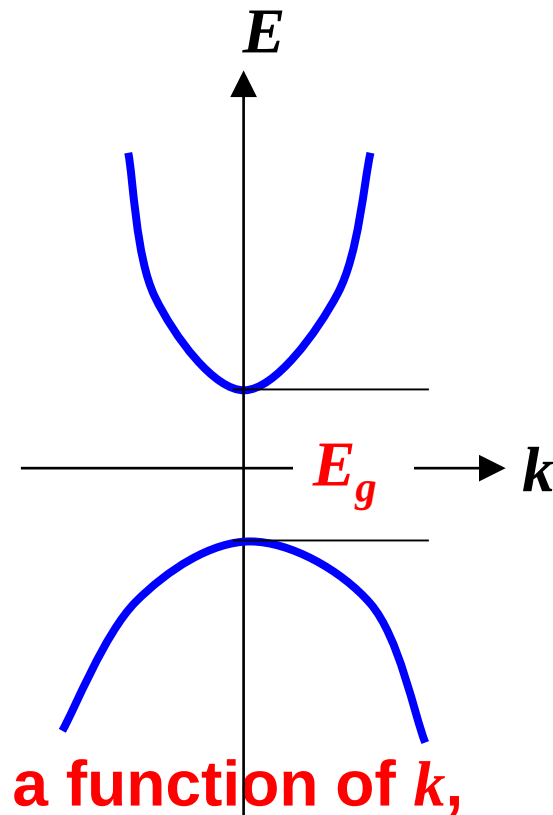
effective mass

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$

The mass that an electron "seems" to have in a solid.
It has nothing to do with the free electron mass m_0

For 3D solids, a tensor form

$$\frac{1}{\mathbf{M}^*} = \frac{1}{\hbar^2} \begin{pmatrix} \frac{\partial^2 E}{\partial k_x^2} & \frac{\partial^2 E}{\partial k_x \partial k_y} & \frac{\partial^2 E}{\partial k_x \partial k_z} \\ \frac{\partial^2 E}{\partial k_y \partial k_x} & \frac{\partial^2 E}{\partial k_y^2} & \frac{\partial^2 E}{\partial k_y \partial k_z} \\ \frac{\partial^2 E}{\partial k_z \partial k_x} & \frac{\partial^2 E}{\partial k_z \partial k_y} & \frac{\partial^2 E}{\partial k_z^2} \end{pmatrix}$$



m^* is a function of k ,
can be smaller or
larger than m_0 , even
can be negative

$$m_0 = 9.11 \times 10^{-31} \text{ kg}$$

Effective Mass 有效质量

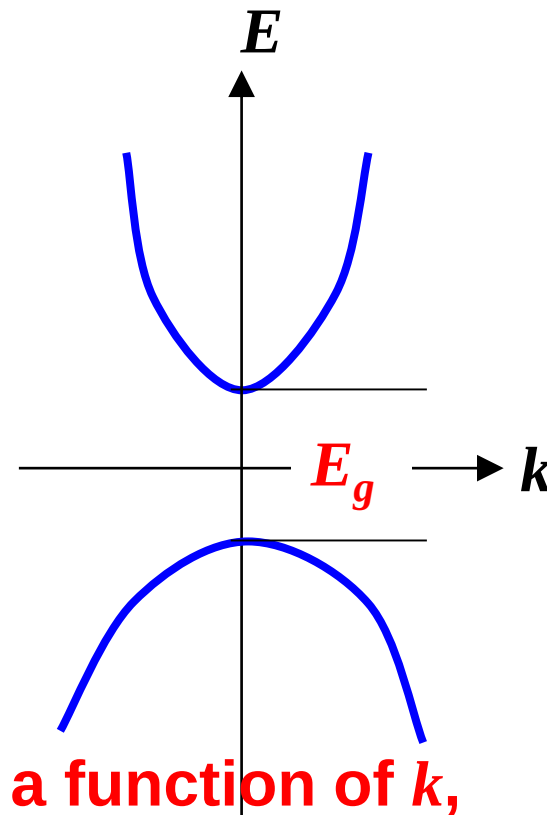
effective mass

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For 3D solids, a tensor form

$$\frac{1}{\mathbf{M}_{ij}^*} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k_i \partial k_j}$$



**m^* is a function of k ,
can be smaller or
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$$m_0 = 9.11 \times 10^{-31} \text{ kg}$$

Effective Mass 有效质量

effective mass

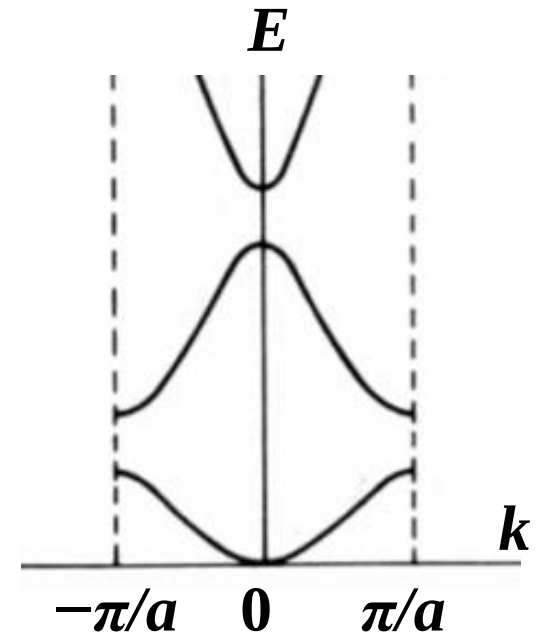
$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$

mobility

$$\mu = \frac{v}{E} = e \frac{\tau}{m^*}$$

conductivity

$$\sigma = ne\mu = ne^2 \frac{\tau}{m^*}$$



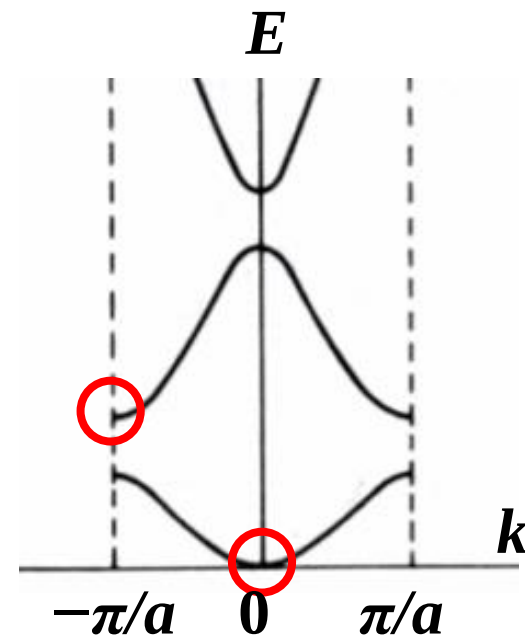
Effective Mass 有效质量

effective mass

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$

close to band minimum
parabolic approximation

$$E(k) \approx E_0 + \frac{\hbar^2}{2m^*} (k - k_0)^2$$



3D DOS

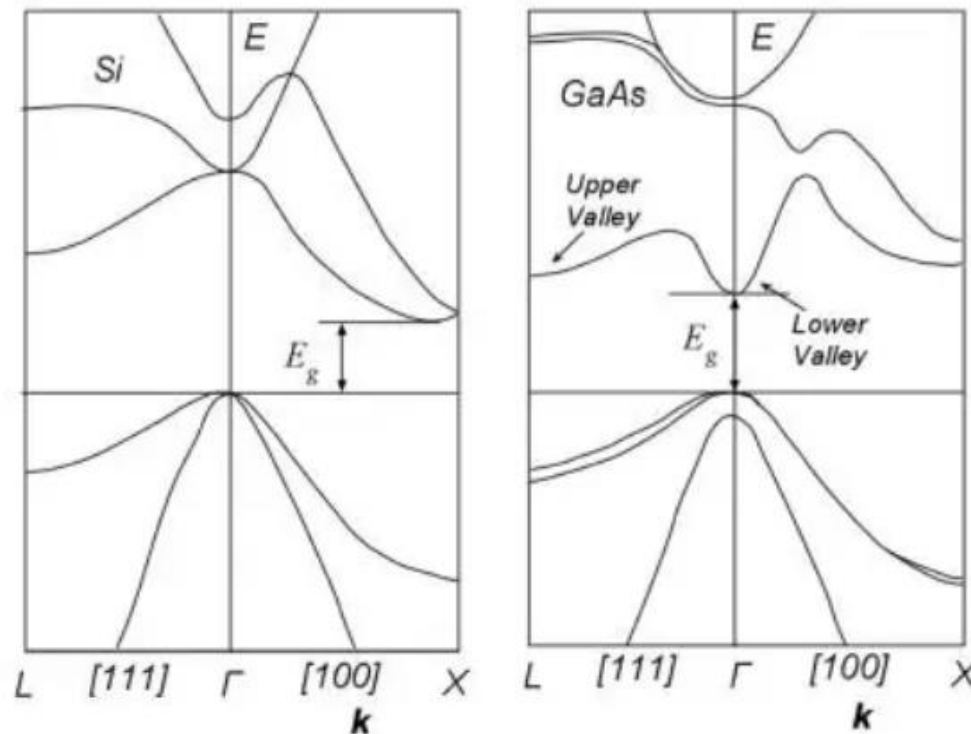
$$g(E) = \frac{dn}{dE} = \frac{1}{2\pi^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} (E - E_0)^{1/2}$$

Effective Mass 有效质量

The actual effective mass is a tensor,
depending on the location (k_x, k_y, k_z)

$$\frac{1}{\mathbf{M}_{ij}^*} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k_i \partial k_j}$$

Approximation is taken for different calculations.



Effective Mass 有效质量

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$$\frac{1}{\mathbf{M}_{ij}^*} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k_i \partial k_j}$$

Approximation is taken for different calculations:

- Density of states calculations

$$g(E) = \frac{dn}{dE} = \frac{1}{2\pi^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} (E - E_0)^{1/2}$$

- Conductivity / mobility calculations

$$\sigma = ne\mu = ne^2 \frac{\tau}{m^*}$$

Effective Mass 有效质量

The actual effective mass is a tensor,
depending on the location (k_x, k_y, k_z)

$$\frac{1}{\mathbf{M}_{ij}^*} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k_i \partial k_j}$$

Approximation is taken for different calculations.

	Symbol	Germanium	Silicon	Gallium Arsenide
Smallest energy bandgap at 300 K	E_g (eV)	0.66	1.12	1.424
Effective mass for density of states calculations				
Electrons	$m_{e^*,dos}/m_0$	0.56	1.08	0.067
Holes	$m_{h^*,dos}/m_0$	0.29	0.57/0.81 ¹	0.47
Effective mass for conductivity calculations				
Electrons	$m_{e^*,cond}/m_0$	0.12	0.26	0.067
Holes	$m_{h^*,cond}/m_0$	0.21	0.36/0.386 ¹	0.34
Free electron mass	m_0 (kg)	9.11×10^{-31}		

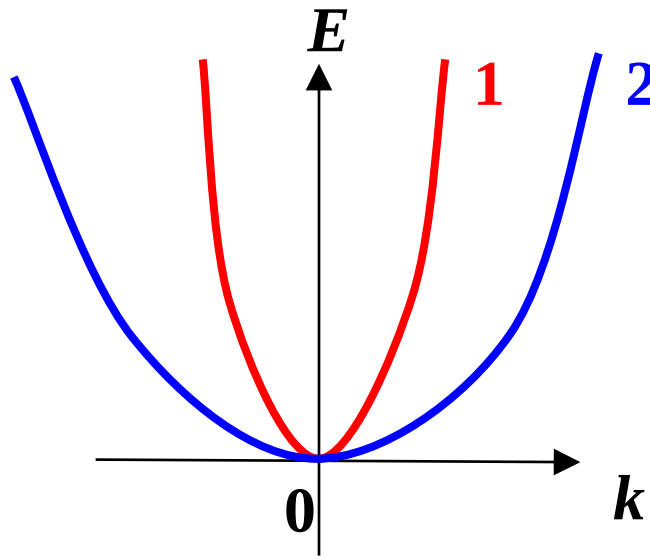
Table 2.3.4 Effective masses for both density of states and conductivity calculations.

Effective Mass 有效质量

effective mass

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$

inverse curvature of the parabolic curve

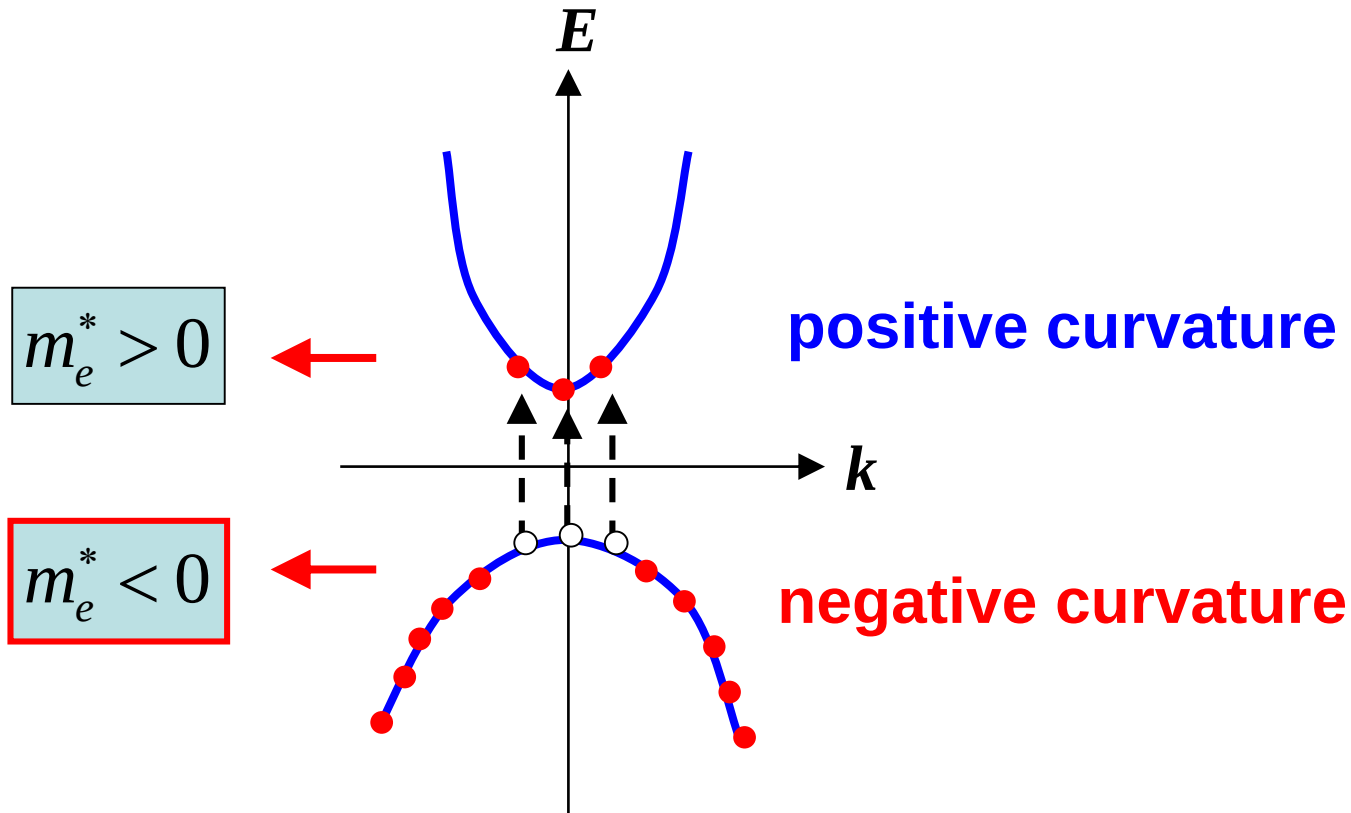


Q: $m_1 > m_2$
or $m_1 < m_2$?

Effective Mass 有效质量

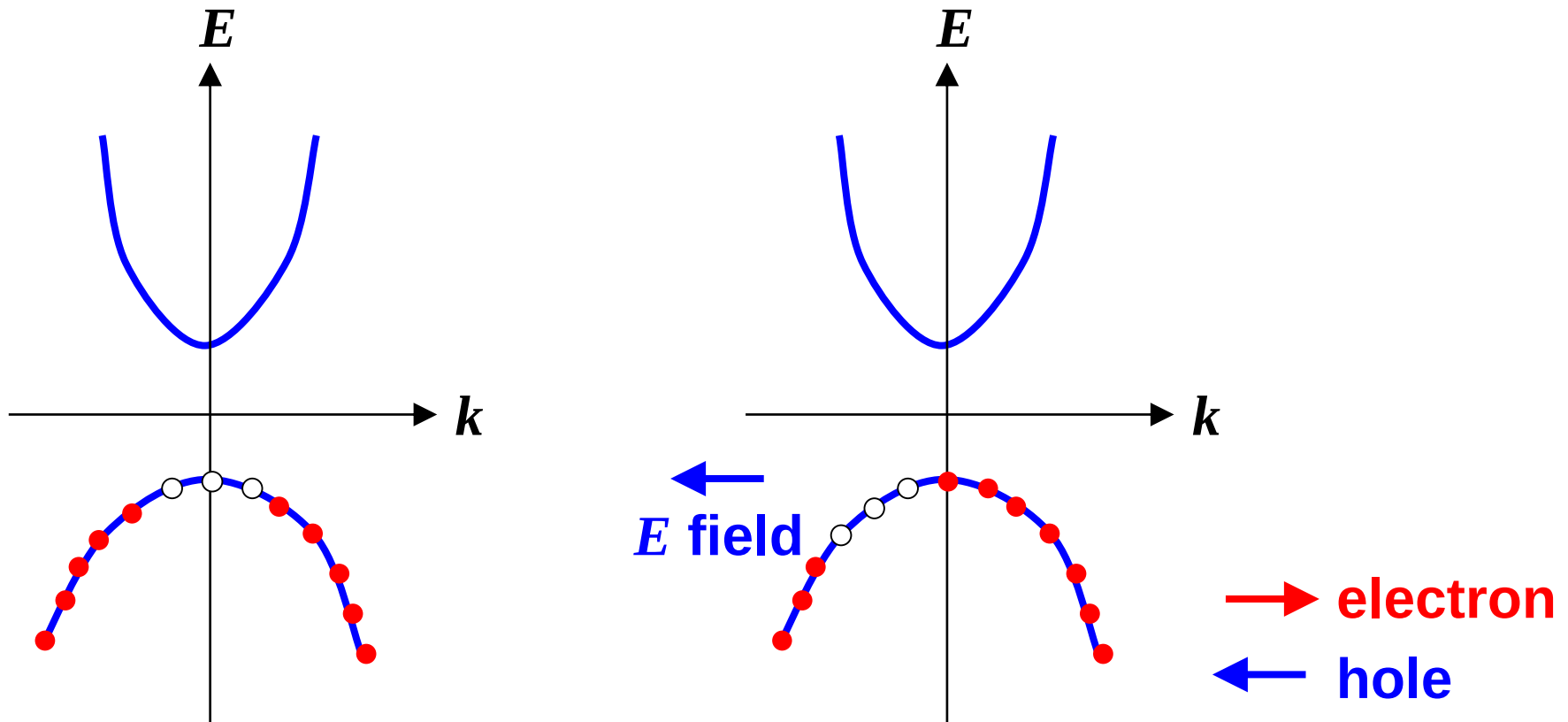
effective mass

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$



Hole 空穴

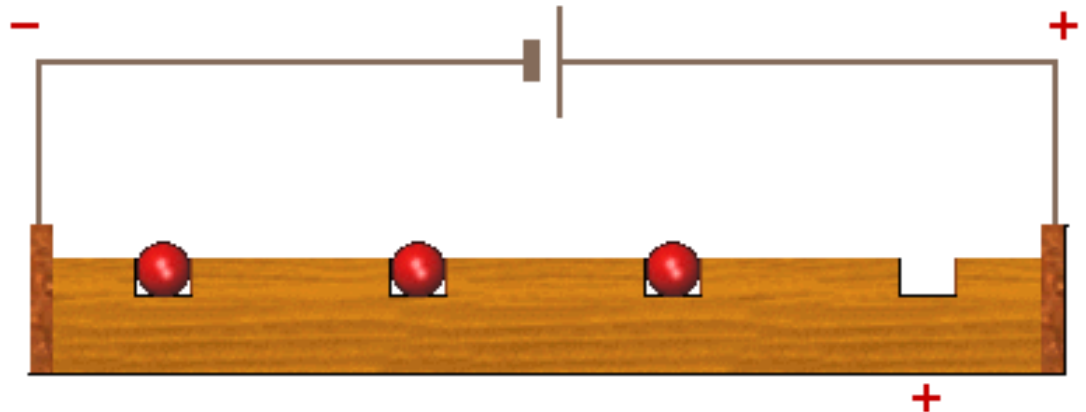
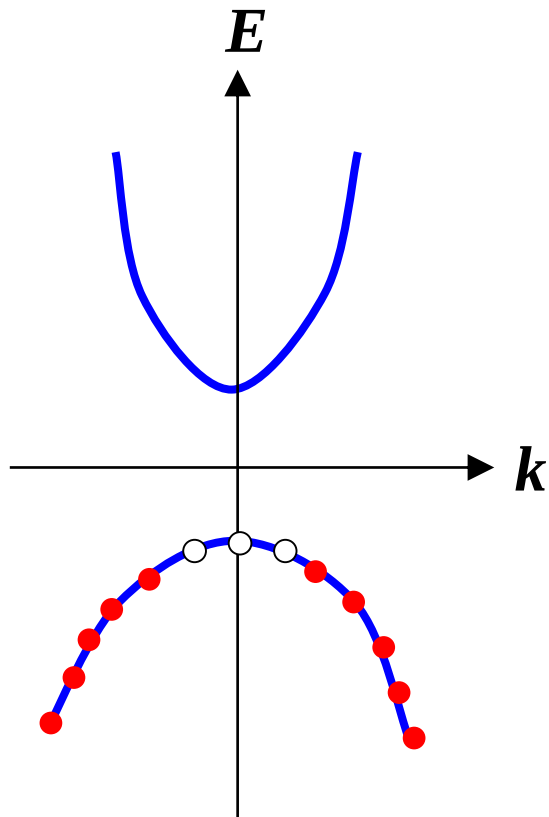
We conventionally use *holes* to analyze the electron behaviors in VB



a hole is the absence of an electron

Hole 空穴

We conventionally use *holes* to analyze the electron behaviors in VB

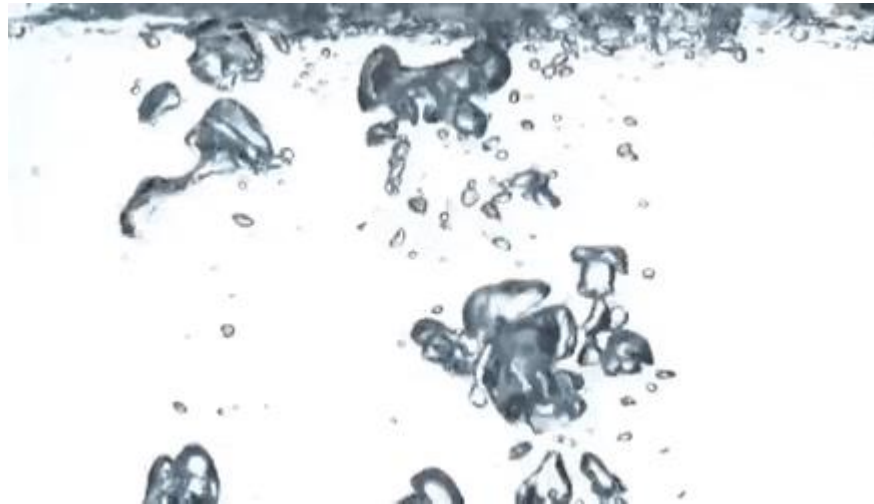
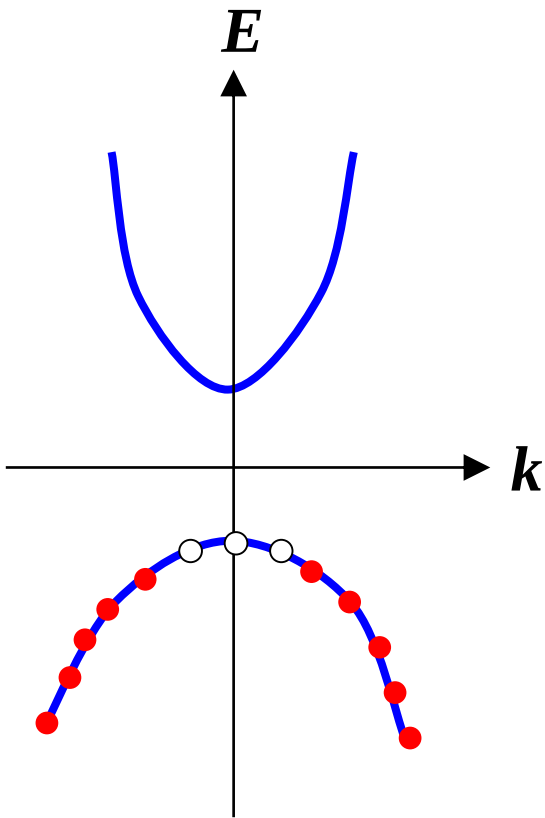


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a hole is a quasi-particle (准粒子),
different from a positron (正电子)

Hole 空穴

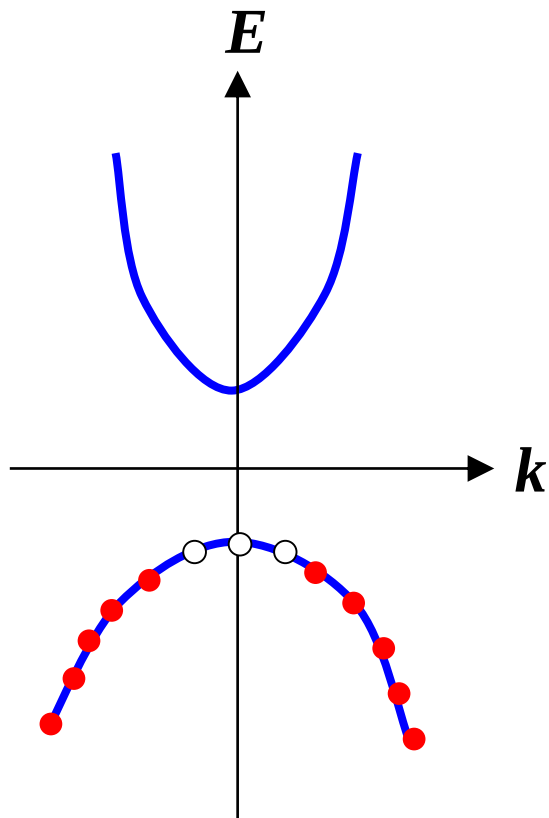
We conventionally use *holes* to analyze the electron behaviors in VB



air bubbles in water

Hole 空穴

In VB, properties of a hole compared to a missing electron in the same position of the band



charge

$$q_h = -e$$

wavevector

$$\mathbf{k}_h = -\mathbf{k}_e$$

energy

$$E_h = -E_e$$

group velocity

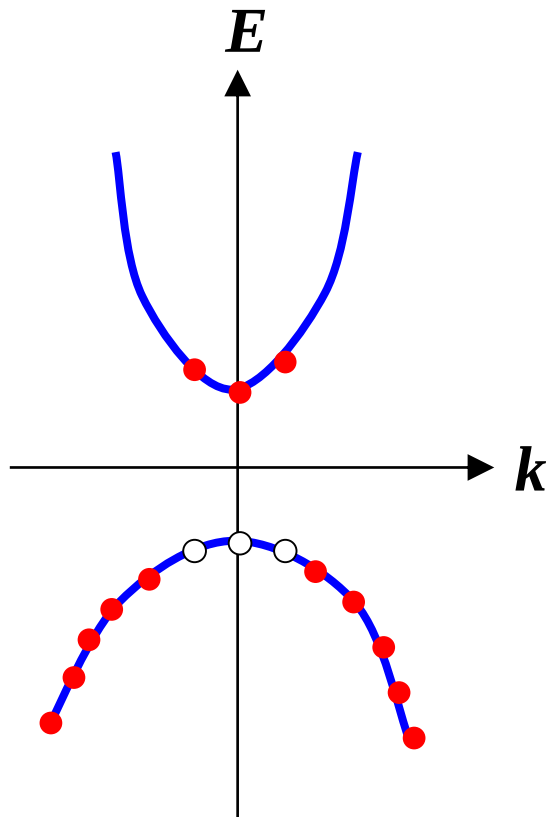
$$\mathbf{v}_h = \mathbf{v}_e$$

effective mass

$$m_h^* = -m_e^*$$

Carriers 载流子

Particles that conduct electrical current:
electrons in CB and holes in VB



in CB

$$m_e^* > 0$$

electron mobility

$$\mu_e = \frac{e\tau}{m_e^*}$$

in VB

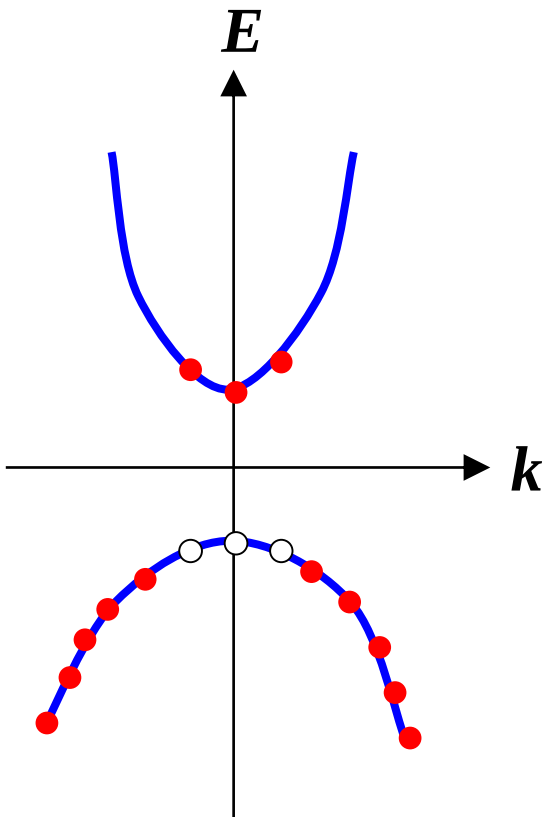
$$m_h^* > 0$$

hole mobility

$$\mu_h = \frac{e\tau}{m_h^*}$$

Carriers 载流子

Particles that conduct electrical current:
electrons in CB and holes in VB



$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$

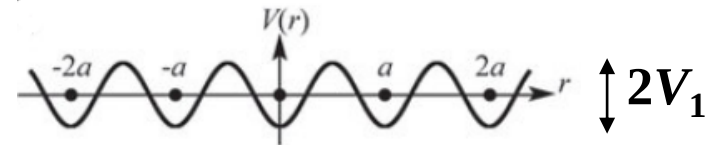
Q: How to calculate:

m_e^* in CB

m_h^* in VB?

The Nearly Free Electron Model

$$\begin{vmatrix} \frac{\hbar^2}{2m_0}(k-g)^2 - E & -V_1 \\ -V_1 & \frac{\hbar^2}{2m_0}k^2 - E \end{vmatrix} = 0$$



$$\rightarrow \left[\frac{\hbar^2}{2m_0}(k-g)^2 - E \right] \left[\frac{\hbar^2}{2m_0}k^2 - E \right] - V_1^2 = 0$$

$$g = \frac{2\pi}{a}$$



$$E_1(k), E_2(k)$$

The Nearly Free Electron Model

Nearly Free electron



$$V_1 \neq 0$$

and

$$V_1 \ll \frac{\hbar^2}{2m_0} \left(\frac{\pi}{a} \right)^2$$

$$\left[\frac{\hbar^2}{2m_0} (k - g)^2 - E \right] \left[\frac{\hbar^2}{2m_0} k^2 - E \right] - V_1^2 = 0$$



$$E_{\pm}(k) = \frac{(A + B) \pm \sqrt{(A - B)^2 + 4V_1^2}}{2}$$

$$A = \frac{\hbar^2}{2m_0} k^2$$

$$B = \frac{\hbar^2}{2m_0} (k - g)^2$$

The Nearly Free Electron Model

Nearly Free electron $\rightarrow$

$$V_1 \neq 0$$

and

$$V_1 \ll \frac{\hbar^2}{2m_0} \left(\frac{\pi}{a} \right)^2$$

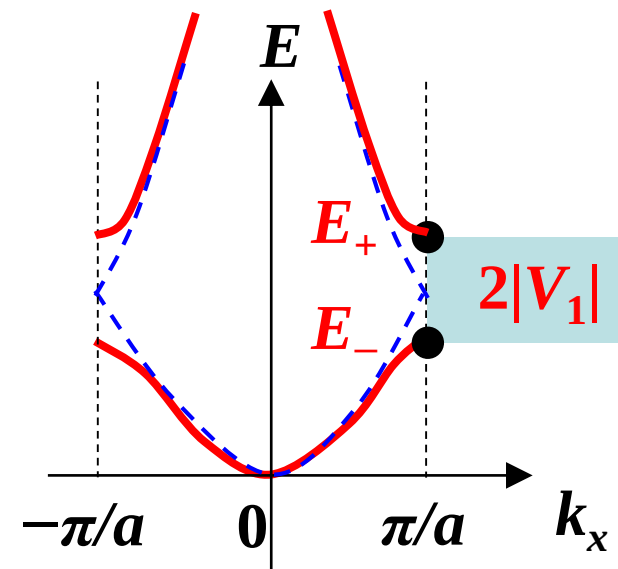
$$\left[\frac{\hbar^2}{2m_0} (k - g)^2 - E \right] \left[\frac{\hbar^2}{2m_0} k^2 - E \right] - V_1^2 = 0$$

when $k = \pi/a$ $\rightarrow$

$$A = B$$

$$E_+ \left(k = \frac{\pi}{a} \right) = \frac{\hbar^2}{2m_0} \left(\frac{\pi}{a} \right)^2 + |V_1|$$

$$E_- \left(k = \frac{\pi}{a} \right) = \frac{\hbar^2}{2m_0} \left(\frac{\pi}{a} \right)^2 - |V_1|$$



Band Gap E_g !

The Nearly Free Electron Model

Nearly Free electron $\rightarrow$ $V_1 \neq 0$

$$\left[\frac{\hbar^2}{2m_0} (k - g)^2 - E \right] \left[\frac{\hbar^2}{2m_0} k^2 - E \right] - V_1^2 = 0$$

when $k \sim \pi/a$, $(A-B) \sim 0$, take the first order approximation

$$E_{\pm}(k) = \frac{(A+B) \pm \sqrt{(A-B)^2 + 4V_1^2}}{2}$$

$$\rightarrow E_{\pm}(k) \approx \frac{A+B}{2} \pm V_1 \left[1 + \frac{1}{2} \frac{(A-B)^2}{4V_1^2} \right]$$

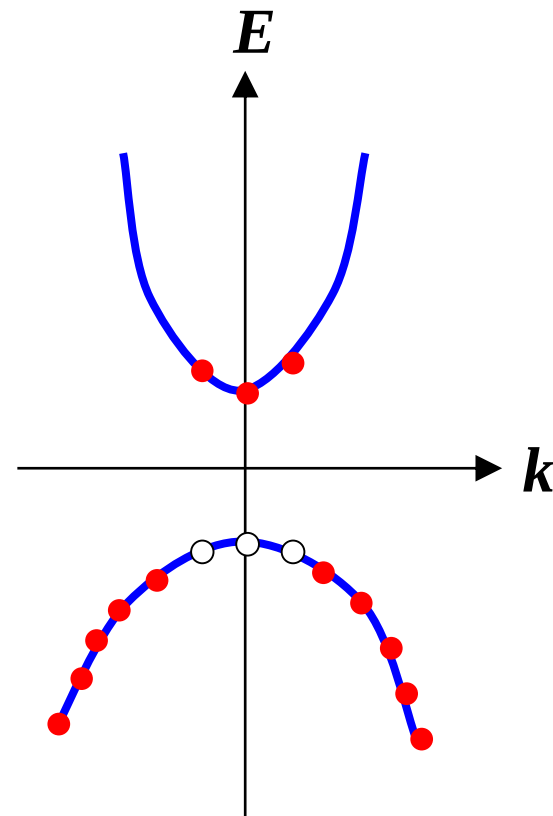
Effective Mass 有效质量

$$E_{\pm}(k) \approx \frac{A+B}{2} \pm V_1 \left[1 + \frac{1}{2} \frac{(A-B)^2}{4V_1^2} \right]$$

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$

→

$$\begin{cases} \frac{m_e^*}{m_0} \approx \frac{1}{C/V_1 + 1} \\ \frac{m_h^*}{m_0} \approx \frac{1}{C/V_1 - 1} \end{cases}$$



$$C = \frac{\hbar^2}{4m_0} g^2 = \frac{\hbar^2}{4m_0} \left(\frac{2\pi}{a} \right)^2 \approx 3 \text{ eV}$$

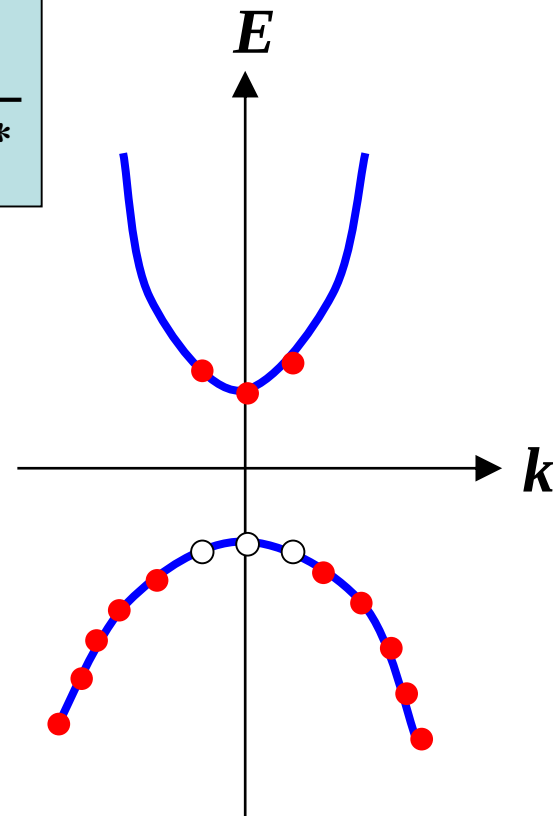
$$V_1 = \frac{1}{2} E_g$$

(assume $a \approx 0.5 \text{ nm}$)

Effective Mass 有效质量

$$\begin{cases} \frac{m_e^*}{m_0} \approx \frac{1}{C/V_1 + 1} \\ \frac{m_h^*}{m_0} \approx \frac{1}{C/V_1 - 1} \end{cases}$$

$$\mu = e \frac{\tau}{m^*}$$

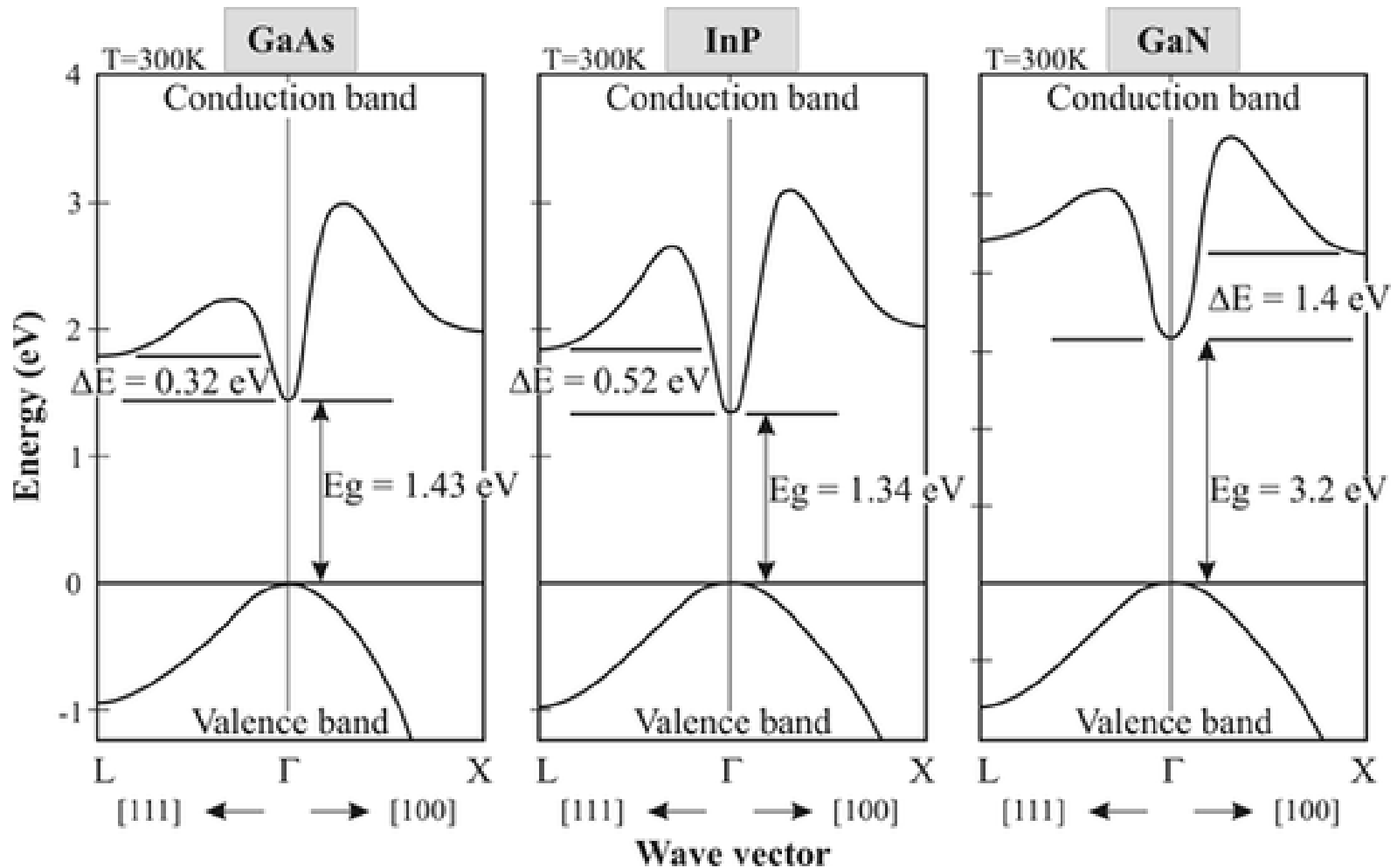


For many semiconductors,
 $C/V_1 \approx 1-10$

→ $m_e^* < m_h^* < m_0$

- Effective mass is smaller than real electron mass m_0
- m_e^* in CB is smaller than m_h^* in VB
- small V_1 ----> small m^* ----> large mobility μ

Examples



m_e^ in CB is smaller than m_h^* in VB
(electrons have more freedom than holes)*

Examples

	a (Å)	E_g (eV)	m_e^* / m_0	m_h^* / m_0	μ_e (cm ² /V/s)	μ_h (cm ² /V/s)
Si	5.43	1.1	0.26	0.38	1350	450
Ge	5.66	0.66	0.12	0.23	3900	1900
-	-	-	-	-	-	-
GaAs	5.65	1.42	0.067	0.45	8500	400
InAs	6.06	0.35	0.022	0.40	33000	450

* effective mass for conductivity

**1. large atoms ----> small V_1 ----> small E_g
 ----> small m^* ----> large mobility μ**

2. $m_e^* < m_h^* < m_0$

3. $\mu_e^* > \mu_h^*$

Examples

	a (Å)	E_g (eV)	m_e^* / m_0	m_h^* / m_0	μ_e (cm ² /V/s)	μ_h (cm ² /V/s)
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* effective mass for conductivity

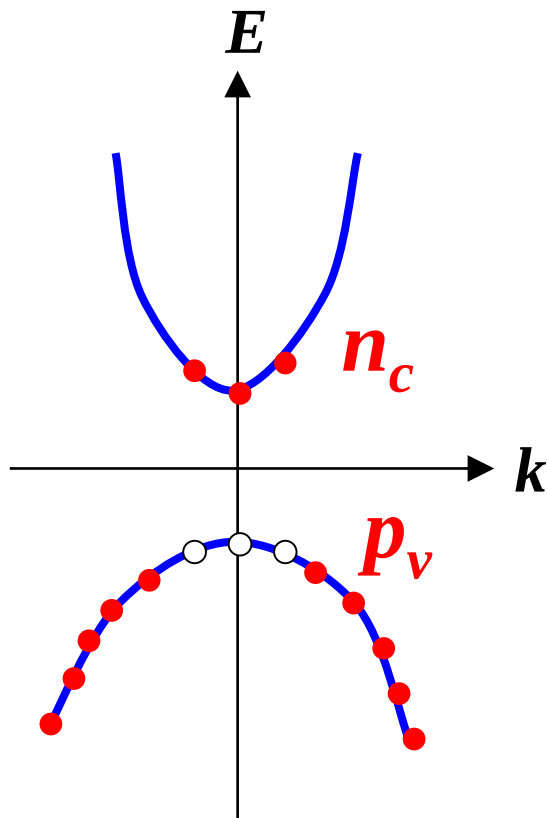
larger atoms

----> electrons have more freedom

----> smaller m^ , move faster*

Carriers 载流子

Particles that conduct electrical current:
electrons in CB and holes in VB



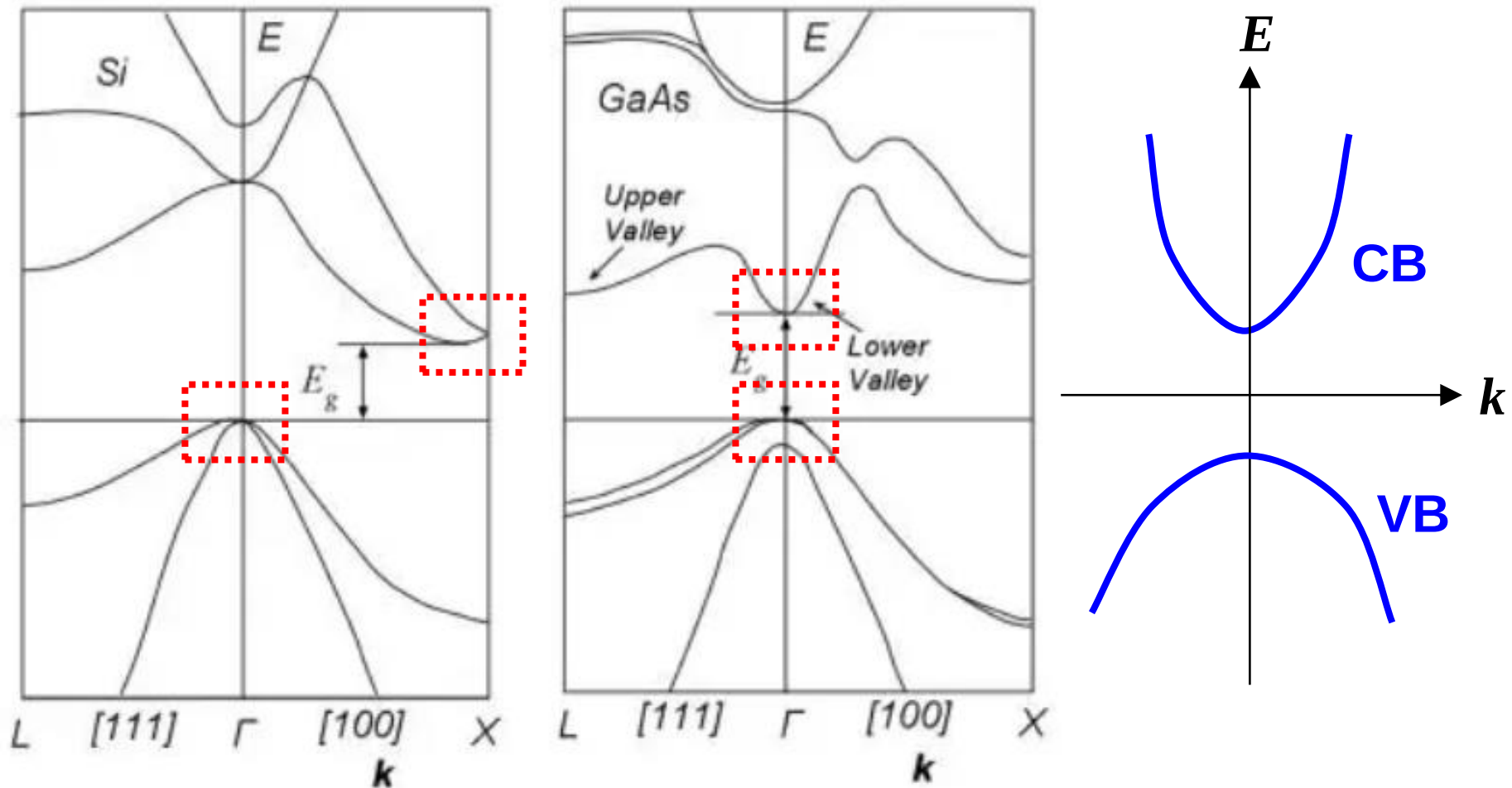
electrical conductivity

$$\sigma = n_c e \mu_e + p_v e \mu_h$$

Q: How to calculate carrier densities?

n_c and p_v (#/cm³)

Band Diagram of Semiconductors



The peaks and valleys of VB and CB can be approximately by *parabolic functions*

Band Diagram of Semiconductors

electrons and holes can be approximated using

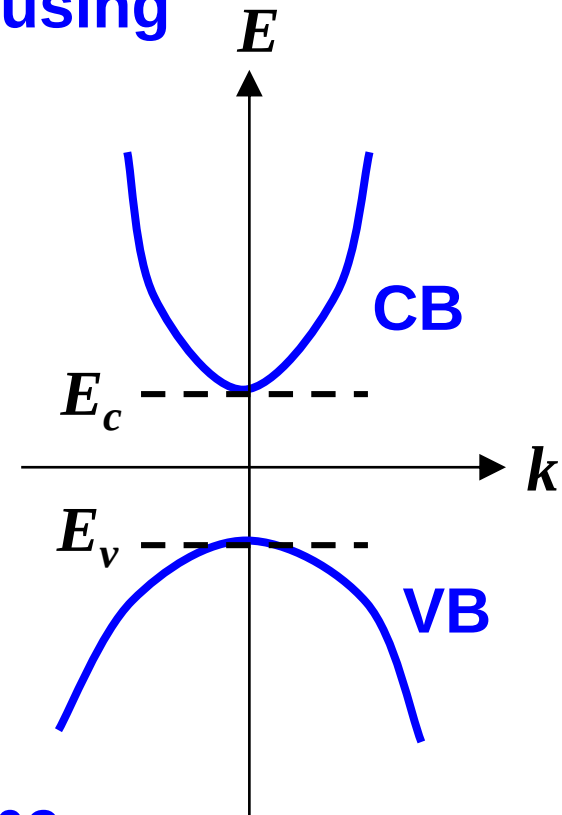
free electron gas

- modify the Sommerfeld Model

$$E_c - E_v = E_g$$

$$E(k) = E_c + \frac{\hbar^2 k^2}{2m_e^*}$$

$$E(k) = E_v - \frac{\hbar^2 k^2}{2m_h^*}$$



The peaks and valleys of VB and CB can be approximated by *parabolic functions*

Q: How many electrons and holes?

Density of States (DOS) 态密度

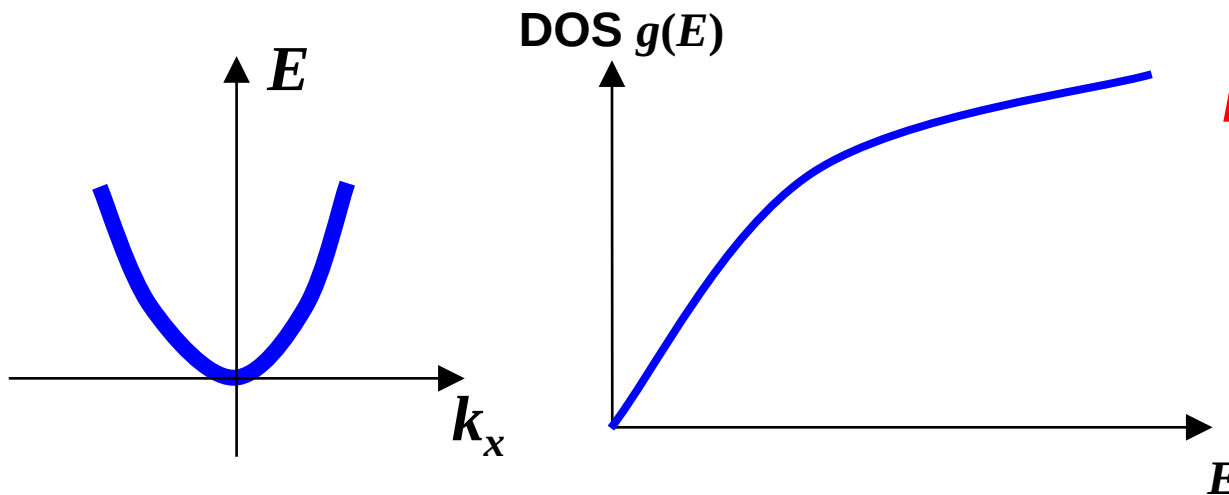
$$g(E) = \frac{dn}{dE}$$

DOS - number of energy states/levels per unit energy in $[E, E+dE]$, per unit volume

**free electrons
in 3D**

$$E = \frac{\hbar^2 k^2}{2m_0}$$

$$g(E) = \frac{dn}{dE} = \frac{1}{2\pi^2} \left(\frac{2m_0}{\hbar^2} \right)^{3/2} E^{1/2}$$



**replace mass m_0
with effective mass m^***

Density of States (DOS) 态密度

$$g(E) = \frac{dn}{dE}$$

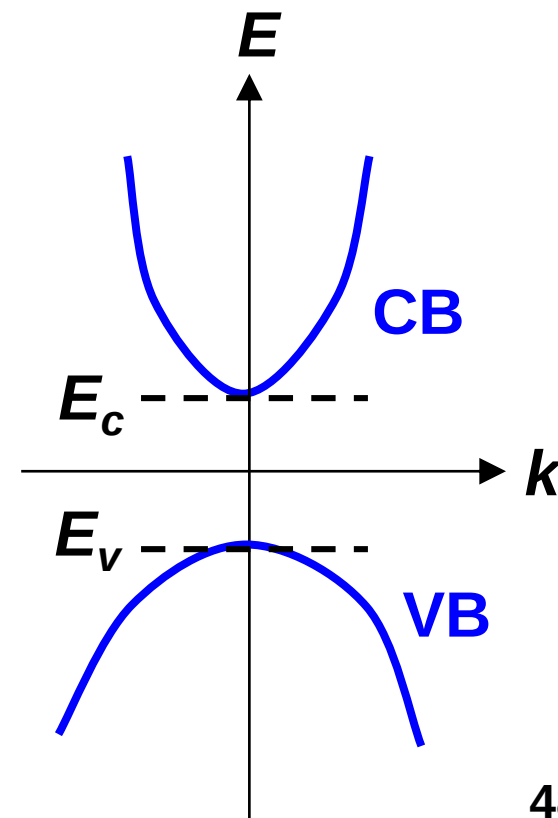
DOS - number of energy states/levels per unit energy in $[E, E+dE]$, per unit volume

DOS for electrons in CB

$$g_c(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} (E - E_c)^{1/2}$$

DOS for holes in VB

$$g_v(E) = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2} \right)^{3/2} (E_v - E)^{1/2}$$



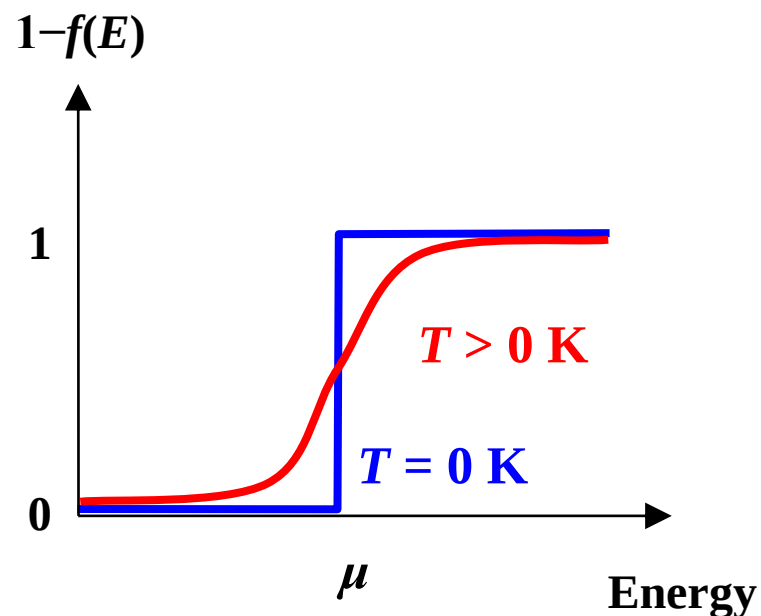
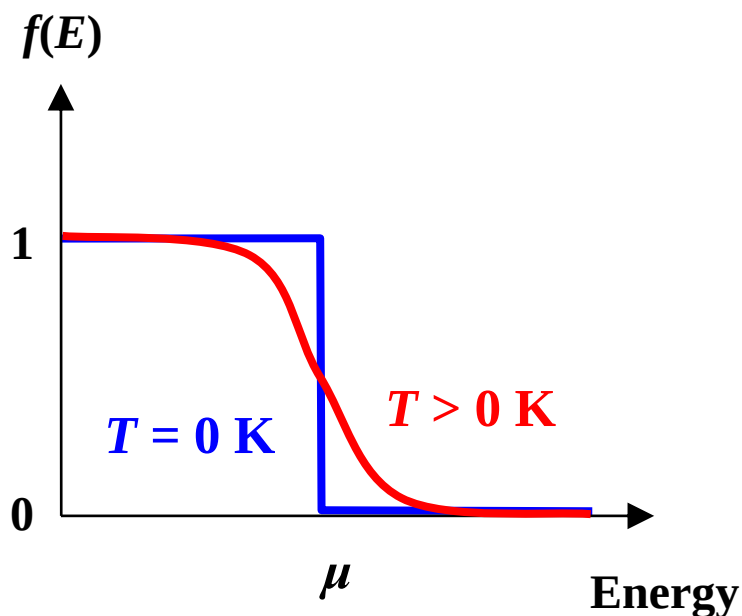
Density of Carriers

**Density of electrons
= DOS * probability f**

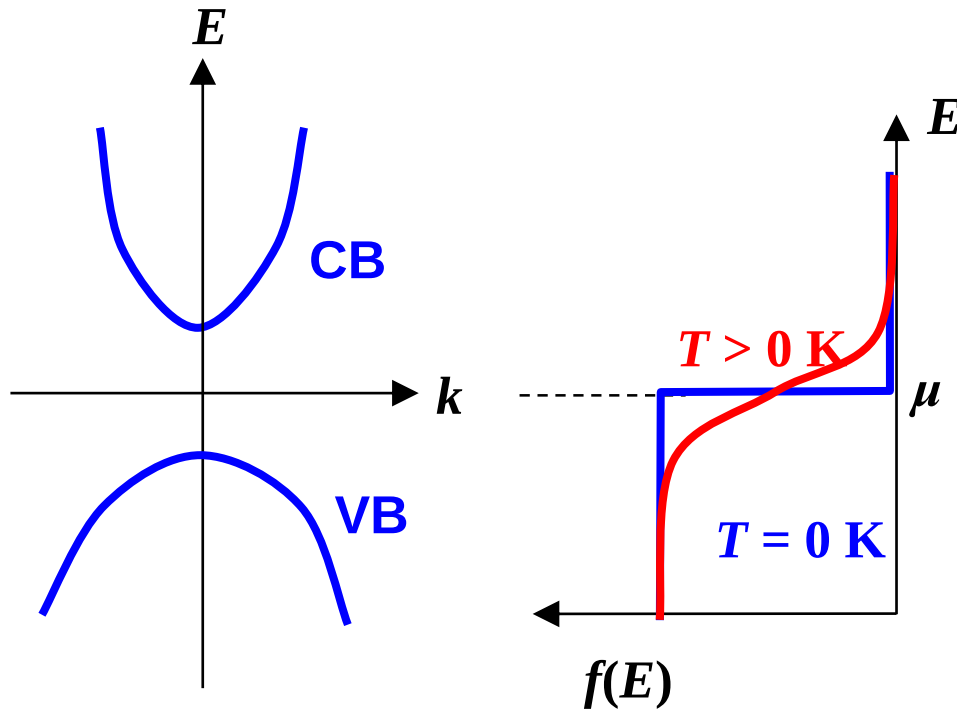
**Density of holes =
DOS * $(1-f)$**

$$f(E) = \frac{1}{e^{(E-\mu)/k_B T} + 1}$$

$$1 - f(E) = 1 - \frac{1}{e^{(E-\mu)/k_B T} + 1} = \frac{1}{e^{(\mu-E)/k_B T} + 1}$$



Chemical Potential μ



For pure semiconductors (intrinsic), the chemical potential μ should lie within the band gap.

Proof: There are very few electrons and holes in intrinsic semiconductors. If μ is too high, there will be too many electrons in CB; Conversely, if μ is too low, there will be too many holes in VB.

Fermi Energy E_F - A Little Note

In metals, Fermi energy/level E_F is the highest occupied state of electrons at $T = 0$ K.

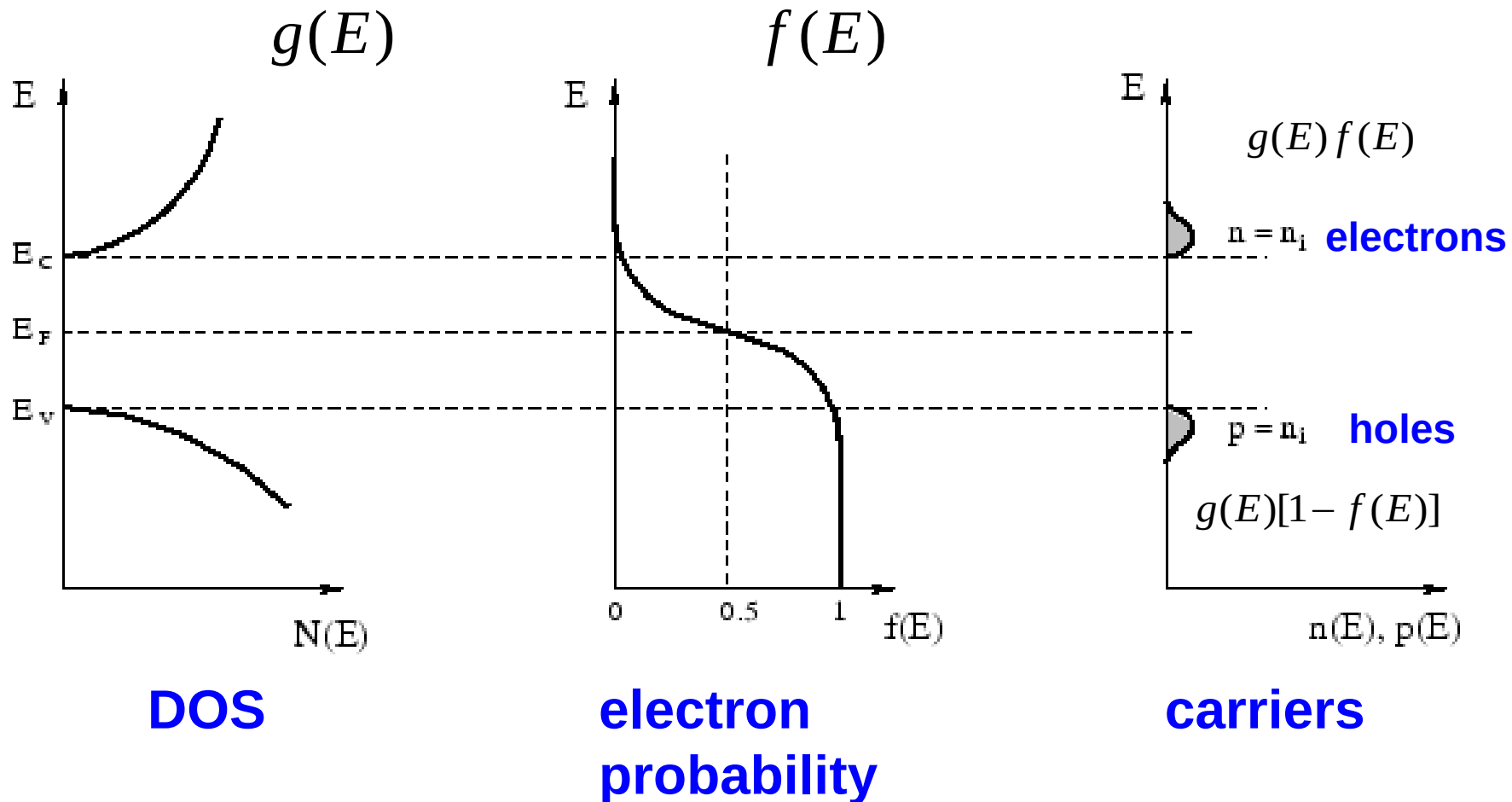
In semiconductors, Fermi energy/level E_F is referred to the chemical potential μ , which is inside the gap. It is not a real energy level !
There are no electrons at E_F !

*"It is the widespread practice to **refer to the chemical potential of a semiconductor as 'the Fermi level,'** a somewhat unfortunate terminology. ... The term 'Fermi level' should be regarded as nothing more than **a synonym for 'chemical potential,'** in the context of semiconductors."*

---- Ashcroft & Mermin, p573

Density of Carriers = DOS * Probability

Intrinsic



Density of Carriers

electrons in CB

$$n_c = \int_{E_c}^{+\infty} g_c(E) \cdot f(E) dE$$

If μ is in the gap, assume

$$E_c - \mu \gg k_B T$$



$$f(E) = \frac{1}{e^{(E-\mu)/k_B T} + 1} \approx e^{-(E-\mu)/k_B T}$$

holes in VB

$$p_v = \int_{-\infty}^{E_v} g_v(E) \cdot [1 - f(E)] dE$$

$$\mu - E_v \gg k_B T$$



$$1 - f(E) = \frac{1}{e^{(\mu-E)/k_B T} + 1} \approx e^{-(\mu-E)/k_B T}$$

Non-Degenerate semiconductors (非简并半导体):
Fermi–Dirac is approximated by Boltzmann distribution

Non-Degenerate semiconductors 非简并半导体

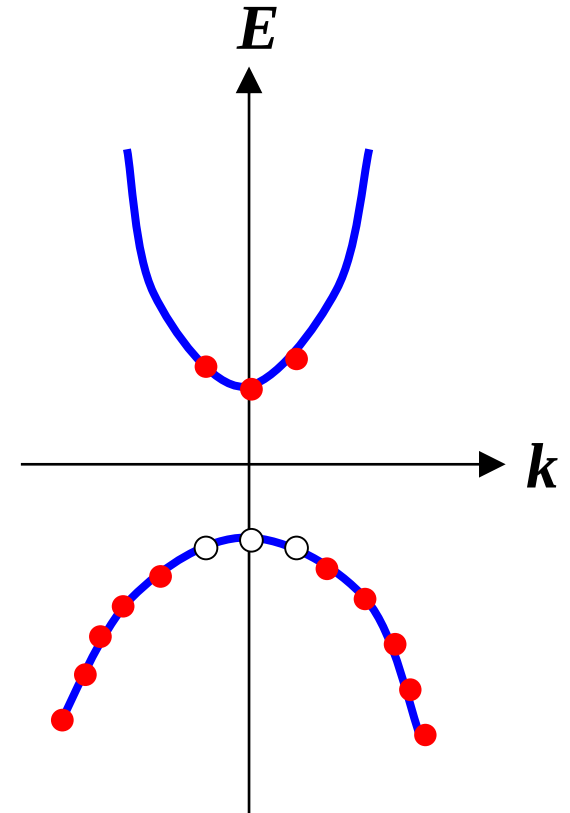
For silicon, the density of k states in VB and CB is approximately the density of Si atoms $\sim 10^{22} \text{ cm}^{-3}$

At room temperature, density of carriers (electrons and holes) is $\sim 10^{10} \text{ cm}^{-3}$ for intrinsic Si, and $\sim 10^{16} - 10^{19} \text{ cm}^{-3}$ for doped Si.

Therefore, the carriers is much fewer than the k states. So non-degeneration approximation is valid for most cases.

Electrons are near the bottom of CB.

Holes are near the top of VB.



However, the approximation is not valid for highly doped semiconductors or at very high temperatures. In these cases, we have to use Fermi–Dirac distribution.

Density of Carriers

electrons in CB

$$\begin{aligned}
 n_c &= \int_{E_c}^{+\infty} g_c(E) \cdot f(E) dE \\
 &= \int_{E_c}^{+\infty} \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} (E - E_c)^{1/2} \cdot e^{-(E-\mu)/k_B T} dE \\
 &= \frac{1}{4} \left(\frac{2m_e^* k_B T}{\pi \hbar^2} \right)^{3/2} e^{-(E_c - \mu)/k_B T} \\
 &= N_c(T) e^{-(E_c - \mu)/k_B T}
 \end{aligned}$$

$$N_c(T) = \frac{1}{4} \left(\frac{2m_e^* k_B T}{\pi \hbar^2} \right)^{3/2} = 2.5 \left(\frac{m_e^*}{m_0} \right)^{3/2} \left(\frac{T}{300 \text{ K}} \right)^{3/2} \times 10^{19} \text{ cm}^{-3}$$

Density of Carriers

note here we use the integral

$$\int_0^{+\infty} x^{1/2} \cdot e^{-x/a} dx = \frac{\sqrt{\pi}}{2} a^{3/2}$$

so

$$\begin{aligned} & \int_{E_c}^{+\infty} (E - E_c)^{1/2} \cdot e^{-(E-\mu)/k_B T} dE \\ &= \frac{\sqrt{\pi}}{2} (k_B T)^{3/2} e^{-(E_c - \mu)/k_B T} \end{aligned}$$

Density of Carriers

holes in VB

$$\begin{aligned}
 p_v &= \int_{-\infty}^{E_v} g_v(E) \cdot [1 - f(E)] dE \\
 &= \int_{-\infty}^{E_v} \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2} \right)^{3/2} (E_v - E)^{1/2} \cdot e^{-(\mu - E)/k_B T} dE \\
 &= \frac{1}{4} \left(\frac{2m_h^* k_B T}{\pi \hbar^2} \right)^{3/2} e^{-(\mu - E_v)/k_B T} \\
 &= P_v(T) e^{-(\mu - E_v)/k_B T}
 \end{aligned}$$

$$P_v(T) = \frac{1}{4} \left(\frac{2m_h^* k_B T}{\pi \hbar^2} \right)^{3/2} = 2.5 \left(\frac{m_h^*}{m_0} \right)^{3/2} \left(\frac{T}{300 \text{ K}} \right)^{3/2} \times 10^{19} \text{ cm}^{-3}$$

Density of Carriers

$$N_c(T) = \frac{1}{4} \left(\frac{2m_e^* k_B T}{\pi \hbar^2} \right)^{3/2} = 2.5 \left(\frac{m_e^*}{m_0} \right)^{3/2} \left(\frac{T}{300 \text{ K}} \right)^{3/2} \times 10^{19} \text{ cm}^{-3}$$

$$P_v(T) = \frac{1}{4} \left(\frac{2m_h^* k_B T}{\pi \hbar^2} \right)^{3/2} = 2.5 \left(\frac{m_h^*}{m_0} \right)^{3/2} \left(\frac{T}{300 \text{ K}} \right)^{3/2} \times 10^{19} \text{ cm}^{-3}$$

effective density of states (有效态密度)

no physical meaning, just two constants

For silicon, at room temperature ($T = 300 \text{ K}$)

$$N_c(T) = 2.73 \times 10^{19} \text{ cm}^{-3} \quad P_v(T) = 1.10 \times 10^{19} \text{ cm}^{-3}$$

$$m_e^* = 1.06m_0$$

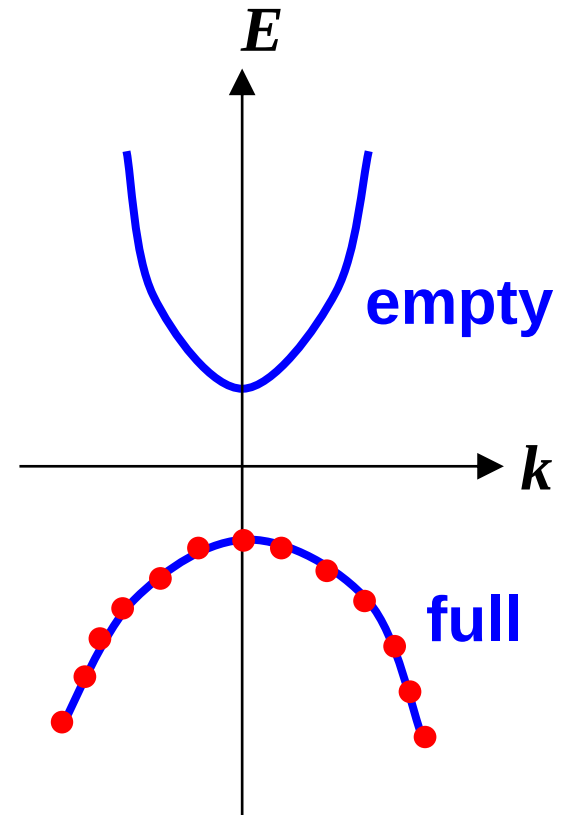
$$m_h^* = 0.58m_0$$

Density of Carriers

when $T = 0$ K

$$n_c = N_c(T)e^{-(E_c - \mu)/k_B T} = 0$$

$$p_v = P_v(T)e^{-(\mu - E_v)/k_B T} = 0$$



$T = 0$ K
insulator

Density of Carriers

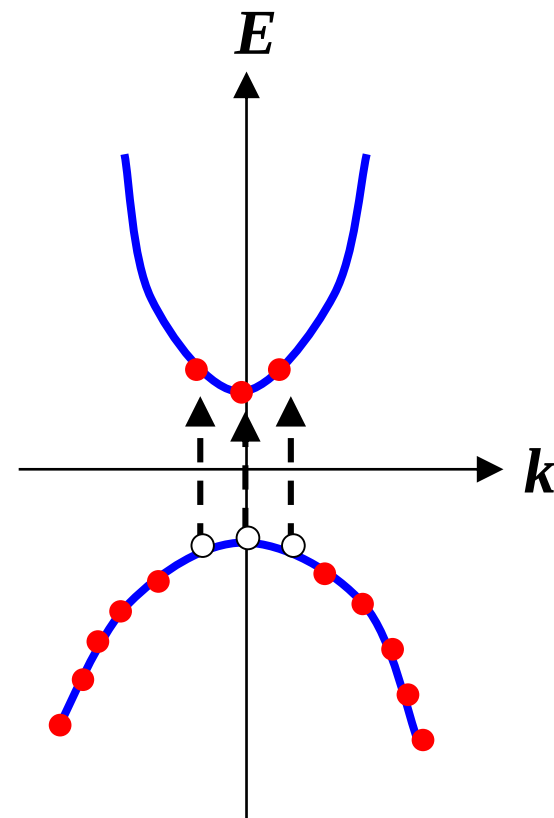
when $T > 0 \text{ K}$

$$n_c = N_c(T) e^{-(E_c - \mu)/k_B T} > 0$$

$$p_v = P_v(T) e^{-(\mu - E_v)/k_B T} > 0$$

conductivity

$$\sigma = n_c e \mu_e + p_v e \mu_h$$



$T > 0 \text{ K}$

thermalization 热激发
CB and VB are partly filled
conductor

Density of Carriers

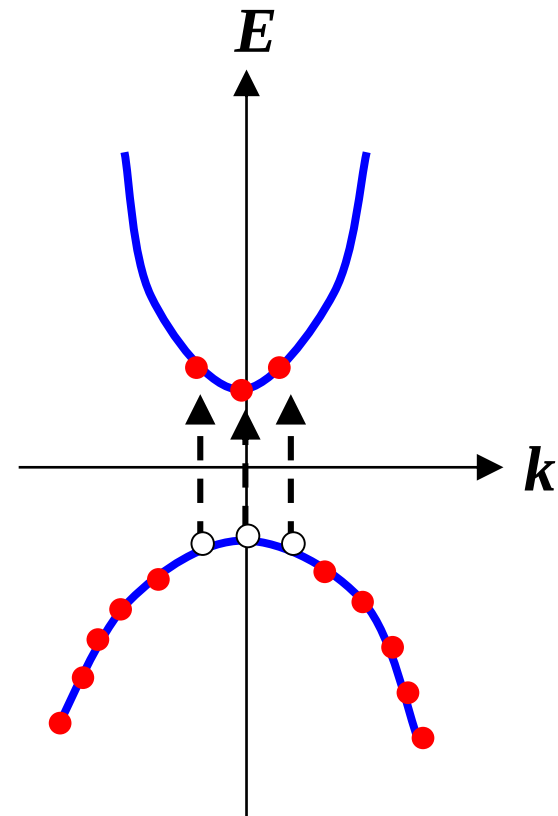
when $T > 0$ K

$$n_c = N_c(T) e^{-(E_c - \mu)/k_B T} > 0$$

$$p_v = P_v(T) e^{-(\mu - E_v)/k_B T} > 0$$



$$\begin{aligned} n_c p_v &= N_c(T) P_v(T) e^{-(E_c - E_v)/k_B T} \\ &= N_c(T) P_v(T) e^{-E_g/k_B T} \end{aligned}$$



mass action law

at equilibrium, $n_c * p_v$ is a constant

Carriers in Semiconductors

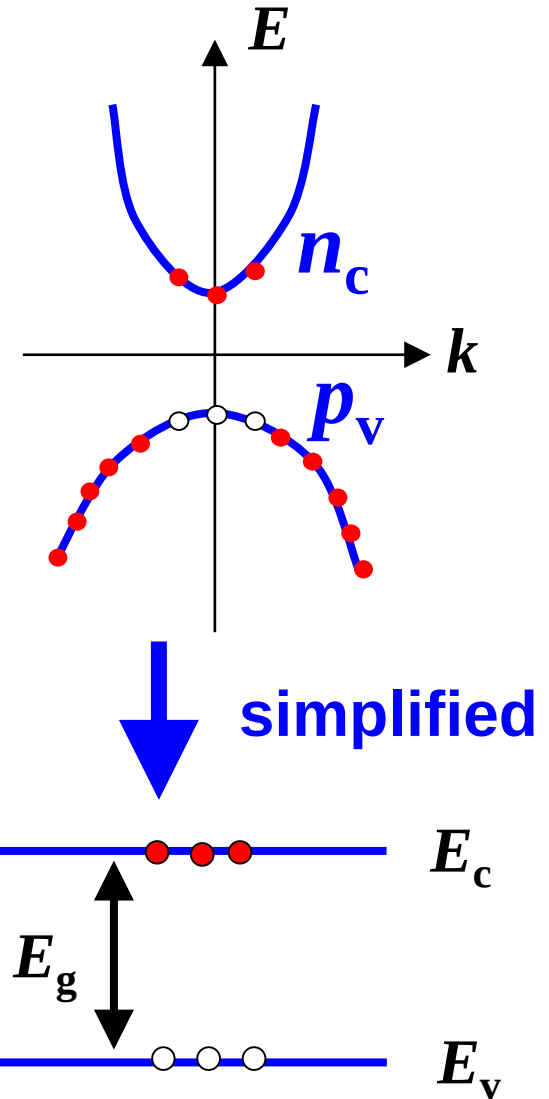
- For non-degenerate semiconductors, we go back to classical physics, assume:

- Carriers are much fewer than DOS

$$f(E) \ll 1$$

- Carriers are non-Degenerate (Boltzmann Distribution)
- Carriers are almost in the same energies (E_c and E_v)
- Carriers have the same velocities and motilities

$$\sigma = n_c e \mu_e + p_v e \mu_h$$



Mass Action Law - A Little Note

- The product of electron and hole concentrations is a constant, at a fixed temperature

$$n_c p_v = n_i^2 = N_c(T) P_v(T) e^{-E_g/k_B T}$$

- In water, the product of H^+ and OH^- concentrations is also a constant

$$[H^+][OH^-] = K_w = 10^{-14} (\text{mol/L})^2 \text{ (at } 25^\circ\text{C)}$$

- Both are originated from classical statistics (non-degenerate, Maxwell–Boltzmann distribution), *not* related to quantum mechanics

Thank you for your attention